EN.553.724 Probabilistic Machine Learning

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Chapter 1

Introduction

1.1 Probabilistic Modeling and Learning

1.1.1 Classical vs. Modern Methods

Classical probabilistic models (e.g. graphical models) and modern "deep" generative models differ primarily in how structure is specified.

- **Classical approaches**
  - Leverage domain knowledge to specify model structure.
  - Work well when the model is correctly specified.
  - Provide interpretable probabilistic quantities and uncertainty quantification.
- **Modern approaches**
  - Learn highly complex distributions directly from data.
  - Require large datasets.
  - Often lack interpretability.
- **Goal**

  combine structured probabilistic modeling with flexible function approximation.

1.1.2 Learning a Function

Model: $y = f(x) + \varepsilon, \quad \mathbb{E}[\varepsilon|x] = 0.$

Given data $\{(x_i, y_i)\}_{i=1}^{N}$, choose f_θ.

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} L(f_\theta(x_i), y_i).$$

1.1.3 Maximum Likelihood Interpretation

Assume: $\varepsilon \sim \mathcal{N}(0, I)$.

Then:

$$p(y_i|x_i, \theta) \propto \exp\left(-\frac{1}{2} \|y_i - f_\theta(x_i)\|^2\right).$$

Proposition 1.1.1. *Minimizing squared loss is equivalent to maximum likelihood estimation.*

$$\arg\max_{\theta} p(y_{1:N}|x_{1:N}) = \arg\min_{\theta} \sum_{i=1}^{N} \|y_i - f_\theta(x_i)\|^2.$$

1.1.4 Learning a Distribution

Given $x_1, \ldots, x_N \overset{\text{iid}}{\sim} \mathbb{P}$:

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{N} \log p_\theta(x_i) = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} -\log p_\theta(x_i).$$

1.2 Measure-Theoretic Foundations

1.2.1 Discrete and Continuous Probability

- Discrete: $\mathbb{P}(A) = \sum_{x \in A} p(x)$
- Continuous: $\mathbb{P}(A) = \int_A p(x)\ \mathrm{d}x$

Unified via: $\mathbb{P}(A) = \int_A p(x)\, \lambda(\mathrm{d}x)$.

1.2.2 Probability Spaces

Definition 1.2.1. A measurable space is $(\Omega, \mathcal{F})$.

Definition 1.2.2. A probability measure $\mathbb{P}$ satisfies:

- $\mathbb{P}(A) \geq 0$
- $\mathbb{P}(\Omega) = 1$
- countable additivity

If a density exists: $\mathbb{P}(A) = \int_A p(x)\, \lambda(\mathrm{d}x)$.

1.2.3 Expectation

$$\mathbb{E}[f] = \int f(x)\,\mathbb{P}(\,\mathrm{d}x)$$

If density exists: $\mathbb{E}[f] = \int f(x)p(x)\,\lambda(\,\mathrm{d}x)$.

1.2.4 Radon–Nikodym Derivative

Definition 1.2.3. $\mathbb{P} \ll Q$ if $Q(A) = 0 \Rightarrow \mathbb{P}(A) = 0$.

Theorem 1.2.1 (Radon–Nikodym)**.** *If* $\mathbb{P} \ll Q$*, then* $\int f\,\mathrm{d}\mathbb{P} = \int f \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}Q}\,\mathrm{d}Q$*.*

If densities exist: $\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}Q}(x) = \frac{p(x)}{q(x)}$.

1.2.5 Random Variables

Definition 1.2.4. A random variable is a measurable function $X : \Omega \to \mathbb{R}$.

Distribution: $\mathbb{P}_X(A) = \mathbb{P}(X \in A)$.

1.2.6 Conditional Probability

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$p(x|y) = \frac{p(x,y)}{p(y)}$$

$$p(x|y) = \frac{p(x)p(y|x)}{p(y)}$$

1.2.7 Beta-Binomial Model

Let M be the number of successes in N Bernoulli trials:

$$\theta \sim \mathrm{Beta}(\alpha, \beta), \qquad M \mid \theta \sim \mathrm{Binomial}(N, \theta).$$

If $M = m$, Bayes' rule gives

$$p(\theta \mid M = m) \propto \theta^{\alpha+m-1}(1-\theta)^{\beta+N-m-1},$$

so $\theta \mid M = m \sim \mathrm{Beta}(\alpha + m, \beta + N - m)$. The posterior mean

$$\mathbb{E}[\theta \mid M = m] = \frac{\alpha + m}{\alpha + \beta + N}$$

is a weighted average of the prior mean and the empirical success rate. The posterior predictive probability of success on the next trial equals this same posterior mean.

1.2.8 Normal–Normal Conjugacy

Let $X_1, \dots, X_n \mid \mu \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with known variance and place the prior $\mu \sim \mathcal{N}(m_0, s_0^2)$. Completing the square in the log posterior gives

$$s_n^{-2} = s_0^{-2} + n\sigma^{-2}, \qquad m_n = s_n^2 \left(s_0^{-2} m_0 + n\sigma^{-2} \overline{X} \right),$$

and hence $\mu \mid X_{1:n} \sim \mathcal{N}(m_n, s_n^2)$. The posterior mean is a precision-weighted average of the prior mean and sample mean.

Integrating over μ gives the posterior predictive distribution

$$X_{n+1} \mid X_{1:n} \sim \mathcal{N}(m_n, \sigma^2 + s_n^2).$$

The variance contains both future observation noise and posterior uncertainty. A plug-in prediction using only $\mathcal{N}(m_n, \sigma^2)$ is too confident.

Example 1.2.1. Take $m_0 = 0$, $s_0^2 = 4$, $\sigma^2 = 1$, $n = 4$, and $\overline{X} = 1.5$. Then $s_n^2 = 4/17$ and $m_n = 24/17 \approx 1.412$. The data dominate because their precision is sixteen times the prior precision.

1.2.9 Conditional Independence

Definition 1.2.5. $X \perp Y | Z$ if $p(x, y|z) = p(x|z)p(y|z)$.

Equivalent formulations, valid almost surely in Z, include

$$p(x \mid y, z) = p(x \mid z)$$

and the conditional-expectation identity

$$\mathbb{E}[f(X)g(Y) \mid Z] = \mathbb{E}[f(X) \mid Z]\mathbb{E}[g(Y) \mid Z]$$

for all bounded measurable f and g. Conditional independence does not in general follow from marginal independence: conditioning can create dependence through a collider, a phenomenon central to graphical models.

1.3 Divergences and Information

1.3.1 Total Variation

Definition 1.3.1. $\mathrm{TV}(\mathbb{P}, Q) = \frac{1}{2} \int |p - q| \, \lambda(\mathrm{d}x)$.

Proposition 1.3.1. $\mathrm{TV}(\mathbb{P}, Q) = \max_A |\mathbb{P}(A) - Q(A)|$.

1.3.2 f-Divergence

Definition 1.3.2. $D_f(\mathbb{P} \| Q) = \mathbb{E}_Q \left[f\left(\frac{p}{q}\right) \right]$.

$$\mathrm{KL}(\mathbb{P} \| Q) = \int p \log \frac{p}{q}.$$

1.3.3 Non-Negativity

Theorem 1.3.1 (Jensen)**.** $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$.

Proposition 1.3.2. $D_f(\mathbb{P}\|Q) \geq 0$.

Proof.

$$\mathbb{E}_Q f\left(\frac{p}{q}\right) \geq f\left(\mathbb{E}_Q \frac{p}{q}\right) = f(1) = 0.$$

□

1.3.4 Entropy and KL

Definition 1.3.3. $H(\mathbb{P}) = -\int p \log p$.

$$\mathrm{KL}(\mathbb{P}\|Q) = -H(\mathbb{P}) - \mathbb{E}_{\mathbb{P}}[\log q].$$

1.3.5 MLE and KL

Proposition 1.3.3. $\mathbb{E}_{\mathbb{P}}[-\log p_\theta(x)] = \mathrm{KL}(\mathbb{P}\|\mathbb{P}_\theta) + H(\mathbb{P})$.

$$\Rightarrow \mathrm{MLE} = \arg\min_\theta \mathrm{KL}(\mathbb{P}\|\mathbb{P}_\theta).$$

1.3.6 KL Direction

$\mathrm{KL}(\mathbb{P}\|Q)$ mass-covering

$\mathrm{KL}(Q\|\mathbb{P})$ mode-seeking

1.3.7 Invariance

Proposition 1.3.4. *For invertible g: $D_f(\mathbb{P}\|Q) = D_f(g_\#\mathbb{P}\|g_\#Q)$.*

Proof. Let $T = \mathrm{d}\mathbb{P}/\,\mathrm{d}Q$. Under an invertible measurable transformation g, the likelihood ratio of the pushforward measures at $y = g(x)$ remains $T(x)$; the Jacobian factors in their two densities cancel. Therefore

$$D_f(g_\#\mathbb{P}\|g_\#Q) = \mathbb{E}_{Y\sim g_\#Q} f(T(g^{-1}(Y))) = \mathbb{E}_{X\sim Q} f(T(X)).$$

□

For a noninvertible measurable map, information can only decrease: $D_f(g_\#\mathbb{P}\|g_\#Q) \leq D_f(\mathbb{P}\|Q)$. This data-processing inequality follows from conditional Jensen applied to the transformed likelihood ratio.

1.4 Optimal Transport and Integral Probability Metrics

1.4.1 Couplings and Wasserstein Distance

Definition 1.4.1. Let $(\Omega, \mathcal{F})$ be a measurable space with metric d, and let μ, ν be probability measures on Ω.

A *coupling* of μ and ν is a probability measure γ on $\Omega \times \Omega$ whose marginals satisfy

$$\gamma(A \times \Omega) = \mu(A), \qquad \gamma(\Omega \times A) = \nu(A).$$

Interpretation: a coupling specifies a joint distribution (X, Y) with marginals $X \sim \mu$, $Y \sim \nu$.

Definition 1.4.2. The Wasserstein-p distance is defined as

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu,\nu)} \mathbb{E}_{(X,Y)\sim\gamma}[d(X,Y)^p] \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of all couplings of μ, ν.

Remark 1.4.1. W_1 is often called the *earth mover's distance*, since it measures the minimal cost of transporting mass from μ to ν.

1.4.2 Integral Probability Metrics

Definition 1.4.3. Let $\mathcal{F}$ be a class of functions $f : \Omega \to \mathbb{R}$. The integral probability metric (IPM) is

$$d_{\mathcal{F}}(\nu, \mu) = \sup_{f \in \mathcal{F}} \left| \mathbb{E}_\nu[f] - \mathbb{E}_\mu[f] \right|.$$

Lemma 1.4.1. *Let* $\mathcal{F} = \{f : \|f\|_\infty \le 1\}$. *Then* $\mathrm{TV}(\mu, \nu) = \frac{1}{2} d_{\mathcal{F}}(\nu, \mu)$.

Proof. For any f with $\|f\|_\infty \le 1$,

$$|\mathbb{E}_\nu[f] - \mathbb{E}_\mu[f]| = \left| \int f(x)(\nu - \mu)(\,\mathrm{d}x) \right| \le \int |f(x)|\,|\nu - \mu|(\,\mathrm{d}x) \le \int |\nu - \mu|(\,\mathrm{d}x).$$

Taking the supremum gives

$$d_{\mathcal{F}}(\nu, \mu) \le 2\,\mathrm{TV}(\mu, \nu).$$

Equality is achieved by choosing $f(x) = \mathrm{sign}(p(x) - q(x))$, which yields

$$\mathbb{E}_\nu[f] - \mathbb{E}_\mu[f] = \int |p(x) - q(x)|\,\mathrm{d}x.$$

□

1.4.3 Kantorovich–Rubinstein Duality

Theorem 1.4.1 (Kantorovich–Rubinstein)**.** *Let $\mathcal{F}$ be the set of 1-Lipschitz functions on $\mathbb{R}^d$. Then $W_1(\mu, \nu) = d_{\mathcal{F}}(\nu, \mu)$.*

Sketch. For any coupling γ and any 1-Lipschitz f,

$$|f(x) - f(y)| \leq d(x, y).$$

Thus

$$\mathbb{E}_\nu[f] - \mathbb{E}_\mu[f] = \mathbb{E}_\gamma[f(Y) - f(X)] \leq \mathbb{E}_\gamma[d(X, Y)].$$

Taking infimum over γ gives

$$d_{\mathcal{F}}(\nu, \mu) \leq W_1(\mu, \nu).$$

The reverse inequality follows by constructing an optimal dual potential. □

Remarks on Metrics vs Divergences

- Total variation and Wasserstein are *metrics*.
- f-divergences (e.g. KL) are not metrics:
  - not symmetric
  - do not satisfy triangle inequality
- IPMs provide a unifying view of distances via function classes.

1.5 Maximum Entropy and Exponential Families

1.5.1 Maximum Entropy Principle

Let P be an unknown distribution on Ω. Suppose we are given the moment constraints $\mathbb{E}_P[\phi_i(X)] = \mu_i$ for $i = 1, \ldots, d$. Equivalently, $\mathbb{E}_P[\phi(X)] = \mu \in \mathbb{R}^d$, where $\phi : \Omega \to \mathbb{R}^d$.

Remark 1.5.1. These constraints alone do not uniquely determine P.

A natural principle is to choose the *least informative* distribution satisfying the constraints.

Definition 1.5.1. With respect to a reference measure ν, the maximum-entropy distribution solves

$$\arg\max_P H_\nu(P) \quad \text{subject to} \quad \mathbb{E}_P[\phi] = \mu, \qquad H_\nu(P) = -\int p(x) \log p(x)\, \nu(\mathrm{d}x).$$

Derivation via Lagrange Multipliers

Assume Ω is finite for simplicity, and write $p(x) = P(X = x)$.

We maximize $-\sum_{x\in\Omega} p(x)\log p(x)$ subject to: $\sum_x p(x) = 1, \quad \sum_x p(x)\phi_i(x) = \mu_i$.

Define the Lagrangian:

$$\mathcal{L}(p, \lambda_0, \lambda) = -\sum_x p(x)\log p(x) + \lambda_0\left(\sum_x p(x) - 1\right) + \sum_{i=1}^{d}\lambda_i\left(\sum_x p(x)\phi_i(x) - \mu_i\right).$$

Taking derivatives:

$$\frac{\partial\mathcal{L}}{\partial p(x)} = -\log p(x) - 1 + \lambda_0 + \sum_{i=1}^{d}\lambda_i\phi_i(x).$$

Setting to zero: $\log p(x) = \lambda_0 - 1 + \sum_{i=1}^{d}\lambda_i\phi_i(x)$.

Thus

$$p(x) \propto \exp\left(\sum_{i=1}^{d}\lambda_i\phi_i(x)\right) = \exp(\langle\theta, \phi(x)\rangle),$$

where $\theta = (\lambda_1, \ldots, \lambda_d)$.

This gives the exponential family form.

1.5.2 Exponential Family

Definition 1.5.2. Let $(\mathcal{X}, \nu)$ be a measure space and $\phi : \mathcal{X} \to \mathbb{R}^d$.

The exponential family is

$$p_\theta(x) = \frac{\exp(\langle\theta, \phi(x)\rangle)}{Z(\theta)} = \exp\Big(\langle\theta, \phi(x)\rangle - A(\theta)\Big),$$

where

$$Z(\theta) = \int \exp(\langle\theta, \phi(x)\rangle)\,\nu(\,\mathrm{d}x), \quad A(\theta) = \log Z(\theta).$$

Remark 1.5.2.

- θ is the natural parameter
- $\phi(x)$ are sufficient statistics
- $A(\theta)$ is the log-partition function

1.5.2.1 Example: Gaussian from Exponential Family

Let $\phi(x) = (x, x^2)$ on $\mathbb{R}$.

Then

$$p_\theta(x) \propto \exp(\theta_1 x + \theta_2 x^2).$$

Completing the square:

$$\theta_1 x + \theta_2 x^2 = \theta_2 \left(x + \frac{\theta_1}{2\theta_2} \right)^2 - \frac{\theta_1^2}{4\theta_2}.$$

For $\theta_2 < 0$, this is a Gaussian:

$$p_\theta(x) = \mathcal{N}(\mu, \sigma^2), \quad \mu = -\frac{\theta_1}{2\theta_2}, \quad \sigma^2 = -\frac{1}{2\theta_2}.$$

1.5.3 Maximum Entropy Characterization

Theorem 1.5.1. *If there exists θ such that $\mathbb{E}_{p_\theta}[\phi(X)] = \mu$, then p_θ is the maximum entropy distribution among all distributions satisfying the moment constraints.*

Proof. Let P satisfy $\mathbb{E}_P[\phi] = \mu$. Then

$$\mathrm{KL}(P \| p_\theta) = \mathbb{E}_P[\log p(x) - \log p_\theta(x)] \geq 0.$$

Rearranging: $-H(P) \geq \mathbb{E}_P[\log p_\theta(x)]$.

Using exponential family form:

$$\mathbb{E}_P[\log p_\theta(x)] = \mathbb{E}_P[\langle \theta, \phi(x) \rangle - A(\theta)] = \langle \theta, \mu \rangle - A(\theta).$$

Thus $H(P) \leq A(\theta) - \langle \theta, \mu \rangle = H(p_\theta)$, so p_θ maximizes entropy. □

1.5.4 Mean Parameter Mapping

Define the mapping

$$\nabla A(\theta) = \mathbb{E}_{p_\theta}[\phi(X)].$$

Definition 1.5.3. Let $\mathcal{M} = \{\mu \in \mathbb{R}^d : \exists P \text{ such that } \mathbb{E}_P[\phi(X)] = \mu\}$ be the set of achievable mean parameters.

Definition 1.5.4. The statistics $\{\phi_i\}_{i=1}^d$ are *minimal* if the functions $1, \phi_1, \ldots, \phi_d$ are linearly independent.

Proposition 1.5.1. *If ϕ is minimal, then:*

- $\nabla A : \Theta \to \mathcal{M}$ *is injective*
- $\nabla A : \Theta \to \mathcal{M}^\circ$ *is surjective (onto the interior)*

Sketch. If ϕ is minimal, then $A(\theta)$ is strictly convex: $\nabla^2 A(\theta) = \mathrm{Cov}_{p_\theta}(\phi(X)) \succ 0$.

Strict convexity implies injectivity of ∇A.

Surjectivity onto the interior follows from convex duality and properties of cumulant generating functions. □

1.5.5 Convexity and Geometry

Proposition 1.5.2. *The log-partition function satisfies:* $\nabla^2 A(\theta) = \mathrm{Cov}_{p_\theta}(\phi(X)) \succeq 0$.

Remark 1.5.3.

- $A(\theta)$ is convex
- strictly convex if ϕ is minimal
- exponential families form a convex dual pair: $\theta \leftrightarrow \mu = \nabla A(\theta)$

Non-Minimal Case

If ϕ is not minimal, then there exists $v \neq 0$ such that

$$\langle v, \phi(x) \rangle = c \quad \forall x.$$

Then $p_{\theta+tv}(x) = p_\theta(x)$, so the parametrization is not identifiable, and ∇A is not injective.

1.6 Conditional Expectation as Projection

Elementary conditional probabilities are insufficient once conditioning events have probability zero. The measure-theoretic object is a conditional expectation.

Definition 1.6.1. For an integrable random variable X and a sub-σ-field $\mathcal{G}$, $\mathbb{E}[X \mid \mathcal{G}]$ is any $\mathcal{G}$-measurable random variable Z satisfying

$$\int_G Z \, \mathrm{d}\mathbb{P} = \int_G X \, \mathrm{d}\mathbb{P} \qquad \text{for every } G \in \mathcal{G}.$$

It exists and is unique up to almost-sure equality.

When $\mathcal{G} = \sigma(Y)$, the Doob–Dynkin lemma provides a measurable function g such that $\mathbb{E}[X \mid \mathcal{G}] = g(Y)$ almost surely. This is the precise meaning of the notation $\mathbb{E}[X \mid Y]$.

Theorem 1.6.1 (L^2 projection)**.** *If $X \in L^2$, then $\mathbb{E}[X \mid \mathcal{G}]$ is the unique minimizer of*

$$\inf_{Z \in L^2(\mathcal{G})} \mathbb{E}(X - Z)^2.$$

Moreover, the residual is orthogonal to every $W \in L^2(\mathcal{G})$:

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])W] = 0.$$

Proof. The defining integral identity first gives orthogonality for indicator functions of events in $\mathcal{G}$, then for simple functions, and finally for all $L^2(\mathcal{G})$ functions by density. For any such Z, expand

$$\mathbb{E}(X - Z)^2 = \mathbb{E}(X - \mathbb{E}[X \mid \mathcal{G}])^2 + \mathbb{E}(\mathbb{E}[X \mid \mathcal{G}] - Z)^2,$$

because the cross term vanishes. □

Important consequences include the tower property $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}X$, the nested tower property, conditional Jensen, and the law of total variance

$$\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y]).$$

Example 1.6.1 (Bayes estimator under squared loss)**.** For any predictor $a(Y)$,

$$\mathbb{E}(X - a(Y))^2 = \mathbb{E}\operatorname{Var}(X \mid Y) + \mathbb{E}(\mathbb{E}[X \mid Y] - a(Y))^2.$$

Thus the posterior mean is the Bayes estimator under squared loss, and its risk is the expected posterior variance.

1.7 KL Chain Rules and Mutual Information

Theorem 1.7.1 (Chain rule for relative entropy)**.** *For joint laws P_{XY} and Q_{XY} with regular conditional distributions,*

$$\mathrm{KL}(P_{XY} \| Q_{XY}) = \mathrm{KL}(P_X \| Q_X) + \mathbb{E}_{X \sim P_X} \mathrm{KL}(P_{Y|X} \| Q_{Y|X}).$$

Proof. Factor the likelihood ratio as

$$\frac{\mathrm{d}P_{XY}}{\mathrm{d}Q_{XY}}(x, y) = \frac{\mathrm{d}P_X}{\mathrm{d}Q_X}(x) \frac{\mathrm{d}P_{Y|X=x}}{\mathrm{d}Q_{Y|X=x}}(y)$$

and take the P_{XY} expectation of its logarithm. □

Mutual information is the divergence from independence:

$$I(X; Y) = \mathrm{KL}(P_{XY} \| P_X P_Y) = \mathbb{E}_X \mathrm{KL}(P_{Y|X} \| P_Y).$$

It is nonnegative, symmetric, and zero exactly when X and Y are independent. The chain rule gives $I(X; Y, Z) = I(X; Y) + I(X; Z \mid Y)$.

Example 1.7.1 (Gaussian mutual information)**.** If (X, Y) is jointly standard Gaussian with correlation ρ, then

$$I(X; Y) = -\frac{1}{2} \log(1 - \rho^2).$$

Information diverges as $|\rho| \uparrow 1$ because one variable becomes an almost deterministic function of the other.

Theorem 1.7.2 (Donsker–Varadhan variational formula)**.** *For $P \ll Q$,*

$$\mathrm{KL}(P \| Q) = \sup_f \left\{ \mathbb{E}_P f - \log \mathbb{E}_Q e^f \right\},$$

where the supremum ranges over functions for which both terms are finite. The optimizer is $f^* = \log(\mathrm{d}P / \mathrm{d}Q)$ up to an additive constant.

Proof. For any f, define the exponentially tilted law $\mathrm{d}Q_f = e^f \,\mathrm{d}Q/\mathbb{E}_Q e^f$. Nonnegativity of $\mathrm{KL}(P\|Q_f)$ rearranges to the upper bound. Choosing the log likelihood ratio attains equality. □

This representation underlies variational mutual-information estimators, energy-based learning, and PAC-Bayesian inequalities. Its empirical use can be unstable because estimating $\log \mathbb{E}_Q e^f$ involves an exponential moment.

Theorem 1.7.3 (Pinsker's inequality)**.** *For probability measures P, Q,*

$$\|P - Q\|_{\mathrm{TV}} \le \sqrt{\frac{1}{2}\mathrm{KL}(P\|Q)}.$$

Pinsker converts likelihood or variational control into expectation bounds for all bounded observables. The reverse implication is false without further conditions: small TV distance does not prevent a likelihood ratio from being extremely large on a rare set.

1.8 Optimal Transport: Examples and Geometry

For $p \ge 1$, the Wasserstein distance is

$$W_p(P,Q) = \left(\inf_{\gamma\in\Gamma(P,Q)} \int \|x-y\|^p \,\mathrm{d}\gamma(x,y)\right)^{1/p}.$$

Unlike KL divergence, it remains finite and informative for many mutually singular distributions. If $P = \delta_x$ and $Q = \delta_y$, then $W_p(P,Q) = \|x-y\|$. This sensitivity to geometry motivates its use in generative modeling.

Example 1.8.1 (One-dimensional quantile coupling)**.** For distributions on $\mathbb{R}$ with quantile functions F^{-1} and G^{-1},

$$W_p(P,Q)^p = \int_0^1 |F^{-1}(u) - G^{-1}(u)|^p \,\mathrm{d}u.$$

The optimal coupling uses a common $U \sim \mathrm{Uniform}(0,1)$ and sets $X = F^{-1}(U)$, $Y = G^{-1}(U)$. Pairing equal quantile ranks avoids crossing transport paths.

Theorem 1.8.1 (Gaussian W_2 distance)**.** *For $P = \mathcal{N}(m_0, C_0)$ and $Q = \mathcal{N}(m_1, C_1)$,*

$$W_2(P,Q)^2 = \|m_0 - m_1\|_2^2 + \mathrm{tr}\left(C_0 + C_1 - 2(C_0^{1/2} C_1 C_0^{1/2})^{1/2}\right).$$

When C_0 and C_1 commute, the covariance term is the squared Frobenius distance between their positive square roots. The noncommutative sandwich in the general formula is essential.

Wasserstein estimation has a serious dimensional cost. For broad classes of d-dimensional distributions, the empirical measure converges in W_p at a rate resembling $n^{-1/d}$ once d dominates $2p$. Structural assumptions, regularization, sliced transport, or learned low-dimensional features are therefore needed in high-dimensional applications.

1.9 Exponential-Family Duality

Consider a minimal regular exponential family

$$p_\theta(x) = h(x)\exp(\langle\theta, \phi(x)\rangle - A(\theta)).$$

Differentiation under the integral gives

$$\nabla A(\theta) = \mathbb{E}_\theta\phi(X), \qquad \nabla^2 A(\theta) = \mathrm{Cov}_\theta(\phi(X)).$$

Minimality makes the covariance positive definite, hence A strictly convex and the natural-to-mean map injective.

The convex conjugate

$$A^*(\mu) = \sup_\theta\{\langle\theta, \mu\rangle - A(\theta)\}$$

is negative entropy up to the contribution of the base measure. At dual points $\mu = \nabla A(\theta)$, $\theta = \nabla A^*(\mu)$ and $A(\theta) + A^*(\mu) = \langle\theta, \mu\rangle$.

Proposition 1.9.1 (KL as a Bregman divergence)**.** *For two members of the same exponential family,*

$$\mathrm{KL}(p_\theta\|p_\eta) = A(\eta) - A(\theta) - \langle\nabla A(\theta), \eta - \theta\rangle.$$

Proof. Take the p_θ expectation of $\log p_\theta(X) - \log p_\eta(X)$ and use $\mathbb{E}_\theta\phi(X) = \nabla A(\theta)$. □

Example 1.9.1 (Bernoulli family)**.** Writing $p = \mathrm{logistic}(\theta)$ gives $A(\theta) = \log(1 + e^\theta)$ and $\nabla A(\theta) = p$. The conjugate is $A^*(p) = p\log p + (1-p)\log(1-p)$, the negative binary entropy. The associated Bregman divergence is Bernoulli KL.

Chapter 2

Graphical Models

2.1 Directed Graphical Models (Bayesian Networks)

2.1.1 Factorization in DAGs

Definition 2.1.1. Let $G = (V, E)$ be a directed acyclic graph (DAG), and let $\{X_v\}_{v \in V}$ be random variables.

A distribution P *factorizes according to* G if

$$p(x_V) = \prod_{v \in V} p\Big(x_v \mid x_{\mathrm{pa}(v)}\Big),$$

where $\mathrm{pa}(v)$ is the set of parents of node v.

Remark 2.1.1. Each factor $p(x_v \mid x_{\mathrm{pa}(v)})$ is a *local conditional model*.

Definition 2.1.2. A *topological ordering* of a DAG is an ordering $(v_1, \ldots, v_n)$ such that $v_i \to v_j \implies i < j$.

Remark 2.1.2. This allows a generative interpretation: $X_{v_1} \to X_{v_2} \to \cdots \to X_{v_n}$, sampling each variable conditioned on its parents.

2.1.2 Independence Structure

Definition 2.1.3. Let $S_v = \Big(\{v\} \cup \mathrm{Desc}(v) \cup \mathrm{pa}(v)\Big)^c$.

A DAG encodes the independence relations $X_v \perp X_{S_v} \mid X_{\mathrm{pa}(v)}$.

Proposition 2.1.1. *A distribution factorizes according to G if and only if it satisfies the independence relations encoded by G.*

2.1.3 Factorization ⇒ Independence (Key Idea)

Lemma 2.1.1. *If P factorizes according to a DAG G, then for any upward-closed set U, $p(x_U) = \prod_{u \in U} p(x_u \mid x_{\mathrm{pa}(u)})$.*

Sketch. Sum out variables not in U using the factorization; terms outside U cancel due to normalization. □

This leads to conditional independence: $X_v \perp X_{S_v} \mid X_{\mathrm{pa}(v)}$.

2.1.4 Constructing a DAG

Given variables $X_1, \ldots, X_n$:

- Choose an ordering $X_1, \ldots, X_n$
- For each k, choose a minimal set $S \subseteq \{1, \ldots, k-1\}$ such that $X_k \perp X_{[k-1]\setminus S} \mid X_S$
- Set $\mathrm{pa}(k) = S$

Remark 2.1.3. The resulting graph is not unique.

Remark 2.1.4. Causality can guide graph construction, but cannot generally be inferred from the graph alone.

2.1.5 Compactness of Representation

For n variables taking values in $\{1, \ldots, r\}$:

- Full joint distribution requires: $r^n - 1$ parameters
- If max indegree is d: $O(nr^{d+1})$ parameters suffice

Remark 2.1.5. Graphical models exploit conditional independence for exponential savings.

2.1.6 Examples

2.1.6.1 Markov Chain

$$X_{k+1} \perp X_1, \ldots, X_{k-1} \mid X_k$$

Factorization: $p(x_1, \ldots, x_n) = p(x_1) \prod_{k=2}^{n} p(x_k \mid x_{k-1})$

2.1.6.2 d-th Order Markov Model

$$X_{k+d} \perp X_1, \ldots, X_{k-1} \mid X_k, \ldots, X_{k+d-1}$$

2.1.6.3 Naive Bayes

Latent class C: $X_k \perp X_{-k} \mid C$

Factorization: $p(c, x_1, \ldots, x_n) = p(c) \prod_{k=1}^{n} p(x_k \mid c)$

2.1.7 Latent Variable Models

Definition 2.1.4. A *latent variable model* introduces hidden variables Z such that $p(x) = \int p(x, z)\ \mathrm{d}z$.

Remark 2.1.6. Latent variables simplify structure by introducing conditional independence.

2.1.7.1 Mixture Model

$$z_i \sim \text{Multinomial}(\pi), \quad x_i \mid z_i \sim \mathcal{N}(\mu_{z_i}, \Sigma_{z_i})$$

2.1.7.2 Stochastic Block Model

$$z_i \sim \text{Multinomial}(\pi), \quad A_{ij} \mid z_i, z_j \sim \text{Bernoulli}(p_{z_i z_j})$$

2.1.7.3 Noisy-OR Model

$$x_j \mid z \sim \text{Bernoulli}\left(1 - \exp\left(-\sum_i w_{ij} z_i\right)\right)$$

2.1.7.4 Hidden Markov Model (HMM)

$$z_t = A z_{t-1} + \epsilon_t, \quad x_t = C z_t + \eta_t$$

Remark 2.1.7. This is a linear Gaussian state-space model.

2.1.8 d-Separation

Definition 2.1.5. A path in a DAG is *blocked* by a set C if one of the following holds:

1. Chain: $X \to Z \to Y$ or $X \leftarrow Z \leftarrow Y$, and $Z \in C$
2. Fork: $X \leftarrow Z \to Y$, and $Z \in C$
3. Collider: $X \to Z \leftarrow Y$, and $Z \notin C$ and no descendant of Z is in C

Definition 2.1.6. A and B are *d-separated* by C if all paths between them are blocked.

Theorem 2.1.1. *If P factorizes according to G and A, B are d-separated by C, then $X_A \perp X_B \mid X_C$.*

2.2 Undirected Graphical Models (Markov Random Fields)

2.2.1 Factorization

Definition 2.2.1. Let $G = (V, E)$ be an undirected graph. Let $\mathcal{C}$ be the set of maximal cliques.

A distribution factorizes over G if $p(x) \propto \exp(-E(x)), \quad E(x) = \sum_{C \in \mathcal{C}} \phi_C(x_C)$, or equivalently $p(x) = \prod_{C \in \mathcal{C}} \psi_C(x_C)$.

Remark 2.2.1. $\psi_C = e^{-\phi_C}$ are called *clique potentials*.

2.2.2 Markov Properties

Definition 2.2.2. P satisfies:

- Global Markov property: $X_A \perp X_B \mid X_C$ if C separates A, B
- Local Markov property: $X_v \perp X_{V \setminus (\{v\} \cup N(v))} \mid X_{N(v)}$
- Pairwise Markov property: $X_v \perp X_w \mid X_{V \setminus \{v,w\}}, \quad (v,w) \notin E$

Remark 2.2.2. Under positivity conditions: global $\iff$ local $\iff$ pairwise

2.2.3 Energy-Based Models

Definition 2.2.3. An energy-based model defines $p_\theta(x) \propto e^{-E_\theta(x)}$.

Remark 2.2.3.

- Lower energy $\Rightarrow$ higher probability
- $E_\theta(x) = \langle \theta, \phi(x) \rangle$ recovers exponential families

2.2.4 Example: Ising Model

$$p(x) \propto \exp\left(\sum_i h_i x_i + \sum_{(i,j) \in E} J_{ij} x_i x_j \right), \quad x_i \in \{-1, 1\}$$

Conditional distribution:

$$p(x_i = 1 \mid x_{-i}) = \frac{\exp(h_i + \sum_{j \in N(i)} J_{ij} x_j)}{\exp(h_i + \sum_{j \in N(i)} J_{ij} x_j) + \exp(-h_i - \sum_{j \in N(i)} J_{ij} x_j)}.$$

Remark 2.2.4. Unlike DAGs, sampling is generally difficult due to the global partition function.

2.3 Additional Properties of Independence

Intersection Property

Proposition 2.3.1. *Suppose $p(x, y, z, w)$ has positive density with respect to a product measure. Then the* intersection property *holds:*

$$\begin{cases} X \perp Y \mid Z, W, \\ X \perp W \mid Z, Y \end{cases} \quad \Rightarrow \quad X \perp (Y, W) \mid Z.$$

Remark 2.3.1. This property does *not* hold in general without positivity.

Proposition 2.3.2. *If the intersection property holds, then the global, local, and pairwise Markov properties are equivalent.*

Hammersley–Clifford Theorem

Theorem 2.3.1 (Hammersley–Clifford)**.** *Let $G = (V, E)$ be an undirected graph. Suppose P has positive density.*

Then:

$$P \text{ satisfies the Markov properties} \iff P \text{ factorizes over } G.$$

Remark 2.3.2. Positivity is essential: without it, independence does not imply factorization.

Canonical Representation

Theorem 2.3.2. *Every strictly positive distribution has the representation*

$$p(x) = \exp\left(\sum_{S \subseteq V} \xi_S(x_S)\right),$$

where

$$\xi_S(x_S) = \sum_{Z \subseteq S} (-1)^{|S|-|Z|} \log p(x_Z, x_{S^c} = 0).$$

Remark 2.3.3.

- This is an inclusion–exclusion type expansion
- If P factorizes over G, then $\xi_S = 0$ for non-cliques S

2.4 Factor Graphs

Definition 2.4.1. A *factor graph* is a bipartite graph with:

- variable nodes V
- factor nodes F

and edges only between variables and factors.

A distribution factorizes as $p(x_V) = \prod_{a \in F} \varphi_a(x_{N(a)})$, where $N(a)$ are variables connected to factor a.

Remark 2.4.1. Factor graphs unify:

- DAG factorizations
- MRF factorizations

2.4.1 Example: Restricted Boltzmann Machine (RBM)

Let visible units $v \in \{0,1\}^{d_v}$ and hidden units $h \in \{0,1\}^{d_h}$.

$$p(v,h) \propto \exp\left(\langle v, Wh\rangle + \langle b, v\rangle + \langle a, h\rangle\right)$$

Remark 2.4.2. Key property: $p(h \mid v) = \prod_j p(h_j \mid v), \quad p(v \mid h) = \prod_i p(v_i \mid h)$

Remark 2.4.3. This conditional independence enables efficient Gibbs sampling.

2.4.2 Example: Coding Theory (LDPC)

$$p(x,z) \propto \prod_i \mathbb{1}\{H_i z = 0\} \prod_j p(x_j \mid z_j)$$

Remark 2.4.4. Inference corresponds to belief propagation.

2.5 Conditional Random Fields

Definition 2.5.1. Let $(V \cup W, E)$ be an undirected graph.

A *conditional random field (CRF)* is a conditional distribution $P(X_W \mid X_V) \propto \prod_{C \in \mathcal{C}} \psi_C(x_C)$, where cliques C involve variables in W.

Remark 2.5.1. CRFs model conditional structure without specifying a joint model over (V, W).

2.6 Directed $\to$ Undirected Conversion

Definition 2.6.1. The *moralized graph* $\mathcal{M}[G]$ of a DAG G is obtained by:

- adding edges between all pairs of parents of each node
- dropping edge directions

Proposition 2.6.1. *If P factorizes according to a DAG G, then it factorizes according to the undirected graph $\mathcal{M}[G]$.*

2.7 Independence via Moralization

Theorem 2.7.1. *Let G be a DAG and A, B, C disjoint sets. The following are equivalent:*

1. *A and B are d-separated by C in G*
2. *A and B are separated by C in the moralized ancestral graph*

Moreover, if P factorizes according to G, then $X_A \perp X_B \mid X_C$.

Definition 2.7.1. The *ancestral graph* is the subgraph induced by A, B, C and their ancestors.

2.8 Markov Blanket

Definition 2.8.1. A set U is a *Markov blanket* for v if $X_v \perp X_{V\setminus(U\cup\{v\})} \mid X_U$.

Proposition 2.8.1. *1. In an undirected graph, $N(v)$ is a Markov blanket.*

2. In a DAG, the Markov blanket is: $\mathrm{pa}(v) \cup \mathrm{ch}(v) \cup \text{co-pa}(v)$.

Remark 2.8.1. The Markov blanket gives the minimal conditioning set needed to isolate a node.

2.9 Learning Graph Structure (Undirected Case)

Proposition 2.9.1. *Assume P is positive. Define graph G by: $(v, w) \in E \iff X_v \not\perp X_w \mid X_{V\setminus\{v,w\}}$.*

Then G is the unique minimal graph for which P satisfies the pairwise Markov property.

At the population level, structure recovery is therefore a collection of conditional-independence decisions. In finite samples these decisions require regularization. For an Ising model

$$p_\Theta(x) \propto \exp\left(\sum_{v<w} \Theta_{vw} x_v x_w + \sum_v h_v x_v\right),$$

the neighborhood of node v can be estimated by ℓ_1-penalized logistic regression of X_v on X_{-v}. Sparsity, a minimum nonzero interaction strength, and an incoherence or restricted-curvature condition are typical assumptions for exact edge recovery. The two nodewise estimates for an edge are reconciled using an AND or OR rule.

Comparison: Directed vs Undirected Models

- Directed models:
  - capture causal / generative structure
  - sampling is straightforward
  - require ordering / domain knowledge
- Undirected models:
  - symmetric relationships
  - easier structure learning

– inference often harder (partition function)

2.10 Inference Problems

2.10.1 Core Tasks

Given a probabilistic model $p_\theta(x)$:

- **Sampling:**
$$x \sim p_\theta(x)$$
- **Marginals:**
$$p_\theta(x_i) = \int p_\theta(x)\, \nu^{\otimes(n-1)}(dx_{-i})$$
- **MAP (maximum a posteriori):**
$$x^* = \arg\max_x p_\theta(x)$$
- **Partition function:** $Z(\theta) = \int \psi_\theta(x)\, \nu^{\otimes n}(dx)$

Remark 2.10.1. These tasks are tightly coupled: computing one often requires solving the others.

Latent Variable Models

For joint models: $p_\theta(x, z) \propto \psi_\theta(x, z)$

posterior inference requires:

$$p_\theta(z|x) = \frac{\psi_\theta(x, z)}{Z(\theta, x)}, \quad Z(\theta, x) = \int \psi_\theta(x, z)\, \nu_Z(dz)$$

- Sampling: $z \sim p_\theta(z|x)$
- Marginals: $p_\theta(z_i|x)$
- MAP: $\arg\max_z p_\theta(z|x)$

2.11 Computational Complexity

Complexity Classes

- $O(n)$, $O(n \log n)$, $O(n^c)$: polynomial time
- $O(2^n)$: exponential time

Definition 2.11.1. A *decision problem* outputs YES/NO.

- **P**: solvable in polynomial time
- **NP**: solutions verifiable in polynomial time
- **NP-complete**: hardest problems in NP
- **NP-hard**: at least as hard as NP-complete

Example 2.11.1. 3-SAT: given a Boolean formula, does a satisfying assignment exist?

Counting Problems

Definition 2.11.2. #P: counting versions of NP problems (e.g., number of satisfying assignments).

Remark 2.11.1. Computing partition functions and marginals is typically #P-hard.

Theorem 2.11.1. *Computing partition functions or marginals in general graphical models is #P-hard.*

Remark 2.11.2. Even approximate counting (e.g., #SAT) can be NP-hard.

2.12 Approximation Algorithms

Definition 2.12.1. A *polynomial-time randomized approximation scheme (PRAS)* for Z outputs, with high probability, an estimate satisfying $e^{-\epsilon}Z \leq \widehat{Z} \leq e^{\epsilon}Z$ in polynomial time.

Definition 2.12.2.

- **FPRAS**: runtime polynomial in n and $1/\epsilon$
- **PTAS**: deterministic approximation scheme

Equivalence: Sampling, Marginals, Partition Function

Definition 2.12.3. A family of problems is *self-reducible* if fixing variables reduces the problem to a smaller instance.

Theorem 2.12.1 (informal)**.** *For distributions $p_\theta(x) \propto \psi_\theta(x)$ on finite spaces:*

1. *Approximate sampling $\Rightarrow$ approximate marginals*
2. *Approximate marginals $\Rightarrow$ approximate sampling (if self-reducible)*
3. *Marginals $\Leftrightarrow$ partition function*

Theorem 2.12.2. *If we can sample within TV distance ϵ in $\mathrm{poly}(n, 1/\epsilon)$ time, then empirical averages approximate the marginals:*

$$\widehat{p}(x_i = a) = \frac{1}{N}\sum_{k=1}^{N} \mathbb{1}\{x_i^{(k)} = a\}.$$

Proof. By Hoeffding's inequality: $\mathbb{P}(|\widehat{p} - p| \geq \delta) \leq 2\exp(-2N\delta^2)$ so $N = O(1/\delta^2)$ suffices. □

2.13 Exact Inference

2.13.1 Variable Elimination

Algorithm 1 Variable Elimination

Input: $p(x) \propto \prod_a \psi_a(x_{\partial a})$
Choose elimination order
for variable i **do**
    Multiply all factors involving i
    Sum out i
    Introduce new factor
end for
Output desired marginal

Remark 2.13.1. Intermediate factors grow in size → exponential complexity.

Example 2.13.1 (Eliminating a three-node chain)**.** Let $p(x_1, x_2, x_3) \propto \psi_{12}(x_1, x_2)\psi_{23}(x_2, x_3)$ and suppose we want $p(x_3)$. Eliminating x_1 first creates the unary factor

$$m_{1\to 2}(x_2) = \sum_{x_1} \psi_{12}(x_1, x_2).$$

Eliminating x_2 then gives

$$\widetilde{p}(x_3) = \sum_{x_2} m_{1\to 2}(x_2)\psi_{23}(x_2, x_3),$$

which is normalized over x_3. No factor ever contains all three variables. Eliminating the center first would create a factor on (x_1, x_3), illustrating why elimination order matters.

Variable elimination is distributivity written as an algorithm. Its cost is controlled by the induced graph: before eliminating a vertex, connect all of its current neighbors, forming a clique, and then remove it. The largest clique created by an order determines the largest intermediate table.

Treewidth

Definition 2.13.1. The *treewidth* $\mathrm{tw}(G)$ is the size of the largest intermediate factor (minus one) in optimal elimination ordering.

Remark 2.13.2. Complexity of variable elimination: $O(n \cdot \mathrm{state}^{\mathrm{tw}(G)+1})$

Finding the minimum-treewidth order is itself NP-hard, so practical systems use heuristics such as minimum degree or minimum fill. A junction tree makes the induced cliques explicit and reuses messages when many marginals are required.

2.14 Belief Propagation

2.14.1 Message Passing

A message summarizes the contribution of the entire subtree on one side of an edge. For a factor graph, variable-to-factor and factor-to-variable messages are

$$m_{i\to a}(x_i) \propto \prod_{b\in\partial i\setminus a} m_{b\to i}(x_i)$$

and

$$m_{a\to i}(x_i) \propto \sum_{x_{\partial a\setminus i}} \psi_a(x_{\partial a}) \prod_{j\in\partial a\setminus i} m_{j\to a}(x_j)$$

. Messages may be normalized at every step because their scale cancels when forming beliefs.

Marginals

$$p(x_i) \propto \prod_{a\in\partial i} m_{a\to i}(x_i)$$

$$p(x_{\partial a}) \propto \psi_a(x_{\partial a}) \prod_{j\in\partial a} m_{j\to a}(x_j)$$

2.14.2 Properties

Theorem 2.14.1 (Exactness of sum-product on trees)**.** *On a tree-structured factor graph, after every directed edge has transmitted one message, the beliefs above equal the exact node and factor marginals. For bounded state size and factor degree, the total cost is linear in the number of graph nodes.*

Proof sketch. Delete an edge (i, a). The tree separates into two components. Induction from the leaves shows that $m_{i\to a}(x_i)$ equals the product of all factors in the component containing i, summed over every variable except the boundary value x_i. The factor-to-variable update performs the analogous sum on the other type of component. Multiplying incoming component contributions at a node reconstructs the full joint distribution summed over all other variables, hence its exact marginal. □

Remark 2.14.1. On loopy graphs:

- becomes *loopy belief propagation*
- no guarantees, but often works well

Example 2.14.1 (Forward–backward as belief propagation)**.** In a hidden Markov model, the message from the past into Z_t is the forward quantity $p(z_t, x_{1:t})$ and the message from the future is $p(x_{t+1:T} \mid z_t)$. Their product, after normalization, is $p(z_t \mid x_{1:T})$. Thus forward–backward is not a separate trick; it is sum-product on a chain.

2.15 Extensions

- Conditioning: add indicator factors
- Continuous variables: approximate messages (e.g., Gaussian)
- Junction tree algorithm: exact inference via tree decomposition

2.16 Takeaways

- Exact inference is exponential in general
- Tree structure enables efficient computation
- Message passing = dynamic programming on graphs
- Sampling, marginals, and partition function are deeply connected

Correctness of Belief Propagation (Trees)

We formalize what the messages compute.

Let $T_{b\to i}$ denote the subtree rooted at b when edge (b, i) is removed.

Proposition 2.16.1. *For a tree-structured factor graph:*

$$Z_{b\to i}(x_i) = \sum_{x_{V(T_{b\to i})\setminus i}} \prod_{c\in F(T_{b\to i})} \psi_c(x_{\partial c}),$$

and similarly

$$Z_{a\to j}(x_j) = \sum_{x_{V(T_{a\to j})\setminus j}} \prod_{c\in F(T_{a\to j})} \psi_c(x_{\partial c}).$$

Proof. By induction on the size of the subtree.

Base case: leaf node. Only one factor contributes, and the expression holds trivially.

Inductive step: assume the claim holds for all subtrees of depth $\leq k$. Then for a larger subtree,

$$Z_{a\to i}(x_i) = \sum_{x_{\partial a\setminus i}} \psi_a(x_{\partial a}) \prod_{j\in\partial a\setminus i} Z_{j\to a}(x_j),$$

which matches the factor-to-variable message update.

Thus the recursion exactly computes subtree marginals. □

Max-Product Belief Propagation (MAP)

Replace sums by maxima.

- Variable $\to$ factor: $M_{i\to a}(x_i) = \prod_{b\in\partial i\setminus a} M_{b\to i}(x_i)$
- Factor $\to$ variable:

$$M_{a\to i}(x_i) = \max_{x_{\partial a\setminus i}} \psi_a(x_{\partial a}) \prod_{j\in\partial a\setminus i} M_{j\to a}(x_j)$$

Remark 2.16.1. This computes max-marginals: $M_i(x_i) \propto \max_{x_{-i}} p(x)$

Algorithm 2 Max-product belief propagation

```
Initialize messages
repeat
    for factor a do
        update M_{a→i}
    end for
    for variable i do
        update M_{i→a}
    end for
until convergence
Output arg max_x p(x) from max-marginals
```

2.16.1 Hidden Markov Models (HMMs)

Model

Latent chain: $z_1 \to z_2 \to \cdots \to z_n, \quad x_t|z_t$

Factorization: $p(z_{1:n}, x_{1:n}) = p(z_1) \prod_{t=2}^n p(z_t|z_{t-1}) \prod_{t=1}^n p(x_t|z_t)$

Forward Pass

Define

$$f_t(i) = p(z_t = i, x_{1:t})$$

Recursion: $f_{t+1}(j) = p(x_{t+1}|z_{t+1} = j) \sum_i f_t(i)\, p(z_{t+1} = j|z_t = i)$

Backward Pass

$$b_t(i) = p(x_{t+1:n}|z_t = i)$$

Recursion: $b_t(i) = \sum_j p(z_{t+1} = j|z_t = i)\, p(x_{t+1}|z_{t+1} = j)\, b_{t+1}(j)$

Posterior

$$p(z_t = i|x_{1:n}) \propto f_t(i)\, b_t(i)$$

Viterbi Algorithm (MAP)

Define

$$\delta_t(j) = \max_{z_{1:t-1}} p(z_{1:t}, x_{1:t})$$

Recursion: $\delta_{t+1}(j) = p(x_{t+1}|z_{t+1} = j) \max_i \delta_t(i)\, p(z_{t+1} = j|z_t = i)$

Remark 2.16.2. Viterbi is max-product BP on a chain.

2.16.2 Loopy Belief Propagation

- Same updates as BP, but on graphs with cycles
- No guarantees of convergence or correctness
- Often works well in practice

Example 2.16.1. LDPC decoding uses loopy BP.

Summary: VE vs BP

- **Variable elimination:**
  - works on any graph
  - computes one marginal at a time
  - complexity depends on treewidth
- **Belief propagation:**
  - exact on trees
  - computes all marginals simultaneously
  - linear-time for trees
  - approximate on loopy graphs

2.17 Takeaways

- Exact inference is exponential in general
- Trees ⇒ efficient dynamic programming
- Message passing unifies many algorithms:

  - forward-backward
  - Viterbi
  - BP
- MAP inference = max-product BP

Chapter 3

Monte Carlo and Markov Chain Sampling

3.1 Sampling Unnormalized Distributions

Suppose the target density is known only up to normalization,

$$\pi(x) = \frac{\widetilde{\pi}(x)}{Z}, \qquad \widetilde{\pi}(x) = e^{-U(x)}, \qquad Z = \int e^{-U(x)}\,\mathrm{d}x.$$

The sampling problem is to construct a random variable with law ν such that $\|\nu - \pi\|_{\mathrm{TV}} \le \varepsilon$, without first computing Z. This situation occurs in Bayesian posterior simulation, Gibbs measures in statistical physics, counting problems, and energy-based generative models.

Once samples are available, integrals become statistical estimation problems. For bounded h,

$$|\mathbb{E}_\nu h - \mathbb{E}_\pi h| \le 2\,\|h\|_\infty\,\|\nu - \pi\|_{\mathrm{TV}}\,,$$

so mixing error translates directly into integration bias. Monte Carlo error and mixing error must both be controlled.

3.2 Monte Carlo Estimation

Given samples $X_1, \dots, X_N \sim p$,

$$\mathbb{E}[f(X)] \approx \frac{1}{N}\sum_{i=1}^{N} f(X_i).$$

Proposition 3.2.1. $\mathrm{Var}\Big(\frac{1}{N}\sum_{i=1}^{N} f(X_i)\Big) = \frac{\mathrm{Var}(f(X))}{N}$.

Proof. Uses independence:

$$\mathrm{Var}\Big(\frac{1}{N}\sum f(X_i)\Big) = \frac{1}{N^2}\sum \mathrm{Var}(f(X_i)) = \frac{1}{N}\mathrm{Var}(f(X)).$$

□

What Distributions Are Easy to Sample?

- Uniform distributions
- Finite discrete distributions
- 1D distributions via inverse CDF
- Product distributions: $p(x_1, \ldots, x_n) = \prod_{i=1}^n p_i(x_i)$
- Directed graphical models: $p(x) = \prod_{i=1}^n p(x_i | x_{\text{parents}(i)})$
- Gaussian distributions

Remark 3.2.1. General high-dimensional distributions are not directly sampleable.

3.2.1 Importance Sampling

Let $Y \sim Q$.

Definition 3.2.1. $\widehat{\mu} = \frac{1}{N} \sum_{i=1}^N \frac{p(Y_i)}{q(Y_i)} f(Y_i)$.

Proposition 3.2.2. $\mathbb{E}_Q[\widehat{\mu}] = \mathbb{E}_P[f(X)]$.

Proof.

$$\mathbb{E}_Q\left[\frac{p(Y)}{q(Y)} f(Y)\right] = \int \frac{p(y)}{q(y)} f(y) q(y)\, dy = \int f(y) p(y)\, dy.$$

□

The support condition $p(x) > 0 \Rightarrow q(x) > 0$ is essential. The estimator has variance

$$\operatorname{Var}_Q\left(\frac{p(Y)}{q(Y)} f(Y)\right) = \int \frac{p(x)^2 f(x)^2}{q(x)}\, \mathrm{d}x - \mu^2,$$

which may be infinite even when f is bounded. A proposal must cover the target tails and should place mass where $|f|p$ is large.

When only unnormalized target values $\widetilde{p}$ are available, use self-normalized importance sampling:

$$\widehat{\mu}_{\text{SN}} = \frac{\sum_{i=1}^N w_i f(Y_i)}{\sum_{i=1}^N w_i}, \qquad w_i = \frac{\widetilde{p}(Y_i)}{q(Y_i)}.$$

This ratio estimator is biased at finite N but consistent under standard moment conditions. A common diagnostic is

$$\text{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2},$$

which equals N for constant weights and approaches one when a single sample dominates. It is a heuristic approximation to variance-based effective sample size, not a universal guarantee.

Example 3.2.1 (Gaussian mean mismatch)**.** Let $P = \mathcal{N}(0, I_d)$ and $Q = \mathcal{N}(m, I_d)$. Then $w(x) = \exp(-m^\top x + \|m\|^2/2)$ under Q, and

$$\mathbb{E}_Q[w^2] = e^{\|m\|^2}.$$

The chi-square divergence and weight variability grow exponentially in the squared separation. If each coordinate is mismatched by a constant amount, $\|m\|^2$ grows linearly with d, demonstrating importance-weight collapse in high dimension.

3.2.2 Rejection Sampling

Algorithm 3 Rejection sampling

Input: $q(x)$, c such that $p(x) \le cq(x)$
repeat
  Sample $Y \sim Q$
  Sample $U \sim \text{Uniform}(0, 1)$
until $U \le \frac{p(Y)}{cq(Y)}$
Output Y

Theorem 3.2.1. *If $p(x) \le cq(x)$, then rejection sampling produces samples from p, and $\#trials \sim \text{Geom}(1/c)$.*

Proof. The probability of accepting a proposal in a measurable set A is

$$\int_A q(y)\frac{p(y)}{cq(y)}\,\mathrm{d}y = \frac{P(A)}{c}.$$

The total acceptance probability is therefore $1/c$, and conditioning on acceptance gives probability $P(A)$. Independent attempts imply the geometric waiting-time distribution. □

Example 3.2.2 (Exponential deterioration in dimension)**.** Let the target be $\mathcal{N}(0, I_d)$ and propose from $\mathcal{N}(0, \sigma^2 I_d)$ with $\sigma > 1$. Then

$$\sup_x \frac{p(x)}{q(x)} = \sigma^d,$$

attained at $x = 0$. The expected number of proposals is σ^d, exponential in dimension. Even a modest one-dimensional mismatch becomes fatal after taking products.

3.3 Markov Chains

Definition 3.3.1. A sequence (X_t) is a Markov chain if $X_{t+1} \perp X_1, \ldots, X_{t-1} \mid X_t$, i.e., $p(x_{t+1}|x_1, \ldots, x_t) = p(x_{t+1}|x_t)$.

Definition 3.3.2. A Markov chain is time-homogeneous if $p(x_{t+1}|x_t)$ does not depend on t.

Example 3.3.1 (Two-state chain)**.** For states $\{0, 1\}$, let

$$K = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \qquad a, b \in (0, 1).$$

Solving $\pi K = \pi$ gives $\pi = (b/(a+b), a/(a+b))$. The second eigenvalue is $1 - a - b$, so deviations from stationarity decay exactly as $|1 - a - b|^t$. This example separates three ideas: stationarity is an algebraic fixed-point condition, convergence is a spectral property, and negative eigenvalues cause oscillation.

Transition Kernels

Definition 3.3.3. A Markov kernel K is a function such that:

- $K(x, \cdot)$ is a probability measure
- measurable in x

For a distribution μ,

$$(\mu K)(A) = \int K(x, A)\, \mu(dx).$$

If densities exist: $(\mu K)(y) = \int \mu(x) k(x, y)\, dx$.

Operator View

Definition 3.3.4. For $f : \Omega \to \mathbb{R}$, $Kf(x) = \mathbb{E}_{Y \sim K(x,\cdot)}[f(Y)] = \int f(y) K(x, dy)$.

Remark 3.3.1. K acts:

- on distributions: $\mu \mapsto \mu K$
- on functions: $f \mapsto Kf$

3.3.1 Stationary Distributions

Definition 3.3.5. π is stationary if $\pi K = \pi$.

Remark 3.3.2. Then $\pi K^t = \pi$ for all t.

3.3.2 Conditions for Convergence

Definition 3.3.6. A Markov chain is:

- irreducible: any state can reach any other
- aperiodic: no deterministic cycles

Theorem 3.3.1. *If a finite Markov chain is irreducible and aperiodic:*

- *it has a unique stationary distribution* π
- $\mu_0 K^t \to \pi$

Theorem 3.3.2 (Ergodic theorem)**.** $\frac{1}{N}\sum_{t=1}^{N} f(X_t) \to \mathbb{E}_\pi[f(X)] \quad a.s..$

3.3.3 Reversibility

Definition 3.3.7. π satisfies detailed balance if $\pi(x)K(x,y) = \pi(y)K(y,x)$.

Proposition 3.3.1. *Detailed balance* $\Rightarrow$ *stationarity.*

Remark 3.3.3. Reversibility makes it easy to design Markov chains.

3.4 Metropolis–Hastings

Target density: $p(x) \propto \tilde{p}(x)$.

Algorithm 4 Metropolis–Hastings

Input: $\tilde{p}(x)$, proposal $q(x,y)$
for $t = 0, \ldots, T$ **do**
Sample proposal $\widehat{X}_{t+1} \sim q(X_t, \cdot)$
Accept with probability

$$\alpha(x,y) = \min\left\{\frac{\tilde{p}(y)q(y,x)}{\tilde{p}(x)q(x,y)}, 1\right\}$$

end for

Theorem 3.4.1. *The MH chain has stationary distribution p.*

Gibbs Sampling

Definition 3.4.1. At each step: $X_i \sim p(x_i|x_{-i})$.

Proposition 3.4.1. *Gibbs sampling is a special case of Metropolis–Hastings with acceptance probability* 1.

Remark 3.4.1. In graphical models: $p(x_i|x_{-i}) = p(x_i|x_{\mathrm{MB}(i)})$.

Why the Gibbs acceptance probability is one. Suppose only coordinate i changes. The proposal density is $q(x,y) = p(y_i \mid x_{-i})$, and $y_{-i} = x_{-i}$. Hence

$$\frac{p(y)q(y,x)}{p(x)q(x,y)} = \frac{p(y_i \mid x_{-i})p(x_{-i})p(x_i \mid x_{-i})}{p(x_i \mid x_{-i})p(x_{-i})p(y_i \mid x_{-i})} = 1.$$

□

Example 3.4.1 (Bivariate Gaussian Gibbs sampler)**.** Let (X_1, X_2) be centered Gaussian with unit marginal variances and correlation ρ. The full conditionals are

$$X_1 \mid X_2 = x_2 \sim \mathcal{N}(\rho x_2, 1-\rho^2), \qquad X_2 \mid X_1 = x_1 \sim \mathcal{N}(\rho x_1, 1-\rho^2).$$

After one complete deterministic scan, $\mathbb{E}[X_2^{(t+1)} \mid X_2^{(t)}] = \rho^2 X_2^{(t)}$. Thus the convergence rate deteriorates as $|\rho| \to 1$: highly correlated coordinates should be updated in blocks or reparameterized.

Example 3.4.2 (Single-site Ising update)**.** For $p(x) \propto \exp(\sum_{(i,j)} J_{ij} x_i x_j + \sum_i h_i x_i)$ with $x_i \in \{-1, 1\}$,

$$\mathbb{P}(X_i = 1 \mid X_{-i}) = \text{logistic}\left(2h_i + 2\sum_{j \in N(i)} J_{ij} X_j\right).$$

Only the Markov blanket $N(i)$ is required. Strong couplings at low temperature create metastable modes, so cheap conditional updates need not imply rapid global mixing.

Examples

Example 3.4.3 (Ball walk)**.** $\widehat{X}_{t+1} \sim \text{Uniform}(B_R(X_t))$, then accept/reject via MH.

Example 3.4.4 (Gibbs sampling)**.** Select coordinate i and sample $X_i \sim p(x_i|x_{-i})$.

Takeaways

- Direct sampling is rarely possible
- MCMC constructs a Markov chain with target stationary distribution
- Convergence requires irreducibility + aperiodicity
- Reversibility simplifies construction (MH)
- Gibbs exploits conditional structure

3.5 Markov Chain Monte Carlo (MCMC)

3.5.1 Sampling via Markov Chains

Definition 3.5.1. Goal: sample from a distribution $p(x) \propto e^{f(x)}, \quad x \in \Omega$, within ε in total variation distance.

Remark 3.5.1. We construct a Markov chain whose stationary distribution is p, then simulate it.

Rejection Sampling (Limitations)

Theorem 3.5.1. *If $p(x) \le cq(x)$, rejection sampling produces samples from p, and #trials $\sim$* $\mathrm{Geom}(1/c)$.

Remark 3.5.2. In high dimensions, c typically grows exponentially, making rejection sampling impractical.

Markov Chains

Definition 3.5.2. A sequence (X_t) is a Markov chain if $p(x_{t+1}|x_1, \dots, x_t) = p(x_{t+1}|x_t)$.

Definition 3.5.3. A distribution π is stationary if $\pi K = \pi$.

3.5.2 Reversibility

Definition 3.5.4. π satisfies detailed balance if $\pi(x)K(x, y) = \pi(y)K(y, x)$.

Proposition 3.5.1. *If detailed balance holds, then π is stationary.*

Proof.

$$\sum_x \pi(x)K(x, y) = \sum_x \pi(y)K(y, x) = \pi(y) \sum_x K(y, x) = \pi(y).$$

□

Metropolis–Hastings

Algorithm 5 Metropolis–Hastings

Input: $\tilde{p}(x)$, proposal $q(x, y)$
for $t = 0, \dots, T$ **do**
  Sample $Y \sim q(X_t, \cdot)$
  Accept with probability

$$\alpha(x, y) = \min\left\{\frac{\tilde{p}(y)q(y, x)}{\tilde{p}(x)q(x, y)}, 1\right\}$$

  Set $X_{t+1} = Y$ if accepted, else $X_{t+1} = X_t$
end for

Theorem 3.5.2. *The Metropolis–Hastings chain has stationary distribution p.*

Proof. Detailed balance follows because

$$p(x)K(x, y) = p(x)q(x, y)\alpha(x, y) = \min\{p(x)q(x, y), p(y)q(y, x)\},$$

and the final expression is symmetric in (x, y). □

Gibbs Sampling

Definition 3.5.5. At each step, pick coordinate i and sample $X_i \sim p(x_i|x_{-i})$.

Proposition 3.5.2. *Gibbs sampling is a special case of Metropolis–Hastings with acceptance probability* 1.

Remark 3.5.3. In graphical models: $p(x_i|x_{-i}) = p(x_i|x_{\mathrm{MB}(i)})$.

3.6 Langevin Monte Carlo (LMC)

Continuous-time process

Definition 3.6.1. Langevin diffusion: $dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dB_t$.

Remark 3.6.1. Target distribution: $p(x) \propto e^{-V(x)}$.

Discretization

Using Euler–Maruyama:

$$X_{t+1} = X_t - \eta \nabla V(X_t) + \sqrt{2\eta}\,\xi_t, \quad \xi_t \sim \mathcal{N}(0, I).$$

Remark 3.6.2. This is noisy gradient descent.

Stationary Distribution

Theorem 3.6.1. *The Langevin diffusion has stationary distribution* $p(x) \propto e^{-V(x)}$.

Proof. From the Fokker–Planck equation: $\frac{\partial \mu_t}{\partial t} = \nabla \cdot (\nabla V\, \mu_t) + \Delta \mu_t$.

Plug $\mu(x) = e^{-V(x)}$:

$$\nabla \mu = -(\nabla V) e^{-V}, \quad \Delta \mu = \left(-\Delta V + \|\nabla V\|^2\right) e^{-V}.$$

Then: $\nabla \cdot (\nabla V \mu) = (\Delta V) e^{-V} - \|\nabla V\|^2 e^{-V}$.

Adding: $\nabla \cdot (\nabla V \mu) + \Delta \mu = 0$.

Thus μ is stationary. □

3.6.1 MALA (Metropolis-adjusted Langevin)

Remark 3.6.3. Corrects discretization bias of LMC.

Algorithm 6 MALA

Propose

$$Y \sim \mathcal{N}(x - h\nabla V(x), 2hI)$$

Accept with MH correction

3.7 Underdamped Langevin Dynamics

Definition 3.7.1. Introduce momentum v:

$$\begin{aligned} dX_t &= v_t\, dt \\ dv_t &= -\gamma v_t\, dt - \nabla V(X_t)\, dt + \sqrt{2\gamma}\, dB_t \end{aligned}$$

Remark 3.7.1. Stationary distribution: $p(x, v) \propto e^{-V(x) - \frac{1}{2}\|v\|^2}$.

Remark 3.7.2. Non-reversible $\to$ faster mixing.

3.8 Hamiltonian Dynamics and Ideal HMC

Hamiltonian dynamics

Introduce an auxiliary momentum p and Hamiltonian $H(q, p) = U(q) + K(p)$. Hamilton's equations are

$$\begin{aligned} \frac{dq}{dt} &= \nabla_p H = \nabla K(p) \\ \frac{dp}{dt} &= -\nabla_q H = -\nabla U(q) \end{aligned}$$

Proposition 3.8.1. *Hamiltonian dynamics preserve the energy $H(q(t), p(t))$ and phase-space volume.*

Proof. Along a trajectory,

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \nabla_q H^\top \nabla_p H - \nabla_p H^\top \nabla_q H = 0.$$

The vector field $(\nabla_p H, -\nabla_q H)$ has zero divergence because the two mixed-derivative traces cancel. Liouville's formula then implies that the flow Jacobian determinant is one. □

Reversibility trick

Define momentum flip: $R(q, p) = (q, -p)$.

Then: $R \circ S_T \circ R = S_T^{-1}$.

Proposition 3.8.2. *HMC transition is reversible w.r.t.*

$$\mu(q, p) \propto e^{-H(q,p)}.$$

Algorithm

Algorithm 7 Hamiltonian Monte Carlo (idealized)

Sample $p \sim e^{-K(p)}$
Simulate Hamiltonian dynamics for time T
Output q

Remark 3.8.1. No rejection needed in exact dynamics (but required after discretization).

Leapfrog Integrator

Algorithm 8 Leapfrog

$p \leftarrow p - \frac{h}{2}\nabla U(q)$
for $t = 1, \dots, T$ **do**
  $q \leftarrow q + h\nabla K(p)$
  $p \leftarrow p - h\nabla U(q)$
end for
$p \leftarrow p - \frac{h}{2}\nabla U(q)$

Remark 3.8.2. Properties:

- volume-preserving
- reversible
- approximately energy-preserving

3.9 Continuous-Time Sampling: Analytic Foundations

The preceding algorithms motivate a more systematic treatment of continuous time. Rather than repeat only their update formulas, this section develops the semigroup, generator, and forward-equation viewpoints used to prove stationarity and convergence.

3.9.1 Overdamped Langevin as a Semigroup

We want to sample from

$$p(x) \propto e^{-V(x)}.$$

Definition 3.9.1. The Langevin diffusion is the stochastic differential equation $dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dB_t$.

Discretizing via Euler–Maruyama:

$$X_{t+1} = X_t - \eta\nabla V(X_t) + \sqrt{2\eta}\,\xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d).$$

Remark 3.9.1. This can be viewed as gradient descent with injected Gaussian noise.

Remark 3.9.2. The continuous-time process has stationary distribution $p(x) \propto e^{-V(x)}$, but the discretized version introduces bias unless η is small.

If V is m-strongly convex and has L-Lipschitz gradient, synchronous coupling gives a direct contraction argument. Drive two solutions with the same Brownian motion. Their difference $D_t = X_t - Y_t$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t} \|D_t\|_2^2 = -2\langle D_t, \nabla V(X_t) - \nabla V(Y_t)\rangle \leq -2m \|D_t\|_2^2 .$$

Therefore $\|D_t\| \leq e^{-mt} \|D_0\|$ and $W_2(\mu P_t, \nu P_t) \leq e^{-mt} W_2(\mu, \nu)$. This proves uniqueness of the invariant law and exponential Wasserstein convergence in the strongly log-concave case.

Example 3.9.1 (Ornstein–Uhlenbeck process)**.** For $V(x) = \frac{1}{2} x^\top H x$ with $H \succ 0$,

$$X_t = e^{-Ht} X_0 + \sqrt{2} \int_0^t e^{-H(t-s)} \, \mathrm{d}B_s.$$

The law is Gaussian with mean $e^{-Ht}\mathbb{E}X_0$ and covariance converging to H^{-1}. Hence the invariant density is proportional to $e^{-x^\top Hx/2}$, and the slowest contraction rate is $\lambda_{\min}(H)$.

3.9.2 Exactness Through Metropolis Adjustment

To correct discretization bias, use Langevin as a proposal in Metropolis–Hastings.

Definition 3.9.2. Proposal distribution: $q(x, y) \propto \exp\left(-\frac{\|y-(x-\eta\nabla V(x))\|^2}{4\eta}\right)$.

Algorithm 9 MALA

Propose $Y \sim \mathcal{N}(x - \eta\nabla V(x),\, 2\eta I)$

Accept with probability

$$\alpha(x, y) = \min\left\{\frac{p(y)q(y, x)}{p(x)q(x, y)}, 1\right\}$$

3.9.3 Kinetic or Underdamped Langevin Dynamics

Definition 3.9.3. Introduce momentum v:

$$\begin{aligned} dX_t &= v_t \, dt, \\ dv_t &= -\gamma v_t \, dt - \nabla V(X_t) \, dt + \sqrt{2\gamma} \, dB_t. \end{aligned}$$

Remark 3.9.3. Stationary distribution: $p(x, v) \propto e^{-V(x) - \frac{1}{2}\|v\|^2}$.

Remark 3.9.4. Noise is injected only in v_t, leading to more efficient exploration.

Although the diffusion matrix is degenerate in (x, v), the velocity noise is transported into position by $\mathrm{d}X_t = V_t\,\mathrm{d}t$. This is the simplest example of a hypoelliptic diffusion. Its generator is

$$\mathcal{L}f = v^\top \nabla_x f - (\gamma v + \nabla V(x))^\top \nabla_v f + \gamma \Delta_v f.$$

Proposition 3.9.1. *The density $\pi(x, v) \propto \exp[-V(x) - \|v\|_2^2/2]$ is invariant for kinetic Langevin dynamics.*

Proof. The adjoint applied to a density ρ is

$$\mathcal{L}^*\rho = -\nabla_x \cdot (v\rho) + \nabla_v \cdot ((\gamma v + \nabla V)\rho) + \gamma \Delta_v \rho.$$

For $\rho = \pi$, the transport terms $v^\top \nabla V \pi$ and $-\nabla V^\top v\pi$ cancel, while the Gaussian friction and velocity-diffusion terms cancel. Thus $\mathcal{L}^*\pi = 0$. □

The process is not reversible under the ordinary time reversal, but it is reversible after flipping velocity. Convergence proofs use *hypocoercivity*: antisymmetric Hamiltonian transport transfers the velocity dissipation to the position coordinates. For ill-conditioned quadratic targets this can improve the dependence on condition number relative to overdamped dynamics.

3.9.4 Continuous-Time Markov Processes

Definition 3.9.4. A continuous-time Markov process $(X_t)_{t\geq 0}$ satisfies $P(X_u \in A | X_{[0,t]}) = P(X_u \in A | X_t), \quad u > t.$

Definition 3.9.5. A family $(P_t)_{t\geq 0}$ is a Markov semigroup if $P_{t+s} = P_t P_s, \quad P_0 = \mathrm{id}.$

The semigroup acts on observables by $P_t f(x) = \mathbb{E}[f(X_t) \mid X_0 = x]$ and dually on initial laws by $\int f\ \mathrm{d}(\mu P_t) = \int P_t f\ \mathrm{d}\mu$. The Chapman–Kolmogorov identity is exactly the semigroup law. Strong continuity, $\|P_t f - f\| \to 0$ as $t \downarrow 0$ in the chosen function space, is what allows an infinitesimal generator to determine the evolution.

3.9.5 Generator of a Markov Process

Definition 3.9.6. The generator is

$$\mathcal{L}f = \lim_{t\downarrow 0} \frac{P_t f - f}{t},$$

for functions f for which this limit exists.

Remark 3.9.5. This defines the process via $P_t = e^{t\mathcal{L}}$.

Proposition 3.9.2 (Itô generator)**.** *If X_t satisfies $\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}B_t$ and $a = \sigma\sigma^\top$, then for $f \in C^2$,*

$$\mathcal{L}f(x) = b(x)^\top \nabla f(x) + \frac{1}{2}\,\mathrm{tr}\Big(a(x)\nabla^2 f(x)\Big).$$

Proof. Itô's formula gives

$$f(X_t) - f(X_0) = \int_0^t \mathcal{L}f(X_s)\,\mathrm{d}s + \int_0^t \nabla f(X_s)^\top \sigma(X_s)\,\mathrm{d}B_s.$$

Condition on $X_0 = x$, take expectations, divide by t, and let t decrease to zero. The stochastic integral has mean zero. □

3.9.6 Fokker–Planck Equation

Let $\mathcal{L}^*$ be the adjoint operator.

Definition 3.9.7. The distribution μ_t evolves as $\frac{d\mu_t}{dt} = \mathcal{L}^*\mu_t$.

If μ_t has density ρ_t, the weak identity

$$\frac{\mathrm{d}}{\mathrm{d}t}\int f\rho_t = \int \mathcal{L}f\,\rho_t = \int f\,\mathcal{L}^*\rho_t$$

defines the adjoint. For an Itô diffusion,

$$\partial_t \rho = -\sum_i \partial_i(b_i\rho) + \frac{1}{2}\sum_{i,j}\partial_{ij}(a_{ij}\rho).$$

This is the Fokker–Planck or Kolmogorov forward equation. The backward equation $\partial_t u = \mathcal{L}u$ instead evolves conditional expectations.

Remark 3.9.6. For Langevin diffusion:

$$\mathcal{L}f = -\langle \nabla f, \nabla V\rangle + \Delta f,$$

$\mathcal{L}^*\mu = \nabla\cdot(\nabla V\mu) + \Delta\mu$.

Stationarity of Langevin

Theorem 3.9.1. *The distribution $\mu(x) \propto e^{-V(x)}$ is stationary for Langevin diffusion.*

Proof. Plug into Fokker–Planck equation:

$$\mathcal{L}^*\mu = \nabla\cdot(\nabla V\mu) + \Delta\mu.$$

Compute: $\nabla\mu = -(\nabla V)\mu, \quad \Delta\mu = (-\Delta V + \|\nabla V\|^2)\mu$.

Then: $\nabla\cdot(\nabla V\mu) = (\Delta V)\mu - \|\nabla V\|^2\mu$.

Adding: $(\Delta V - \|\nabla V\|^2)\mu + (-\Delta V + \|\nabla V\|^2)\mu = 0$. □

3.10 Discretized HMC and Adaptation

3.10.1 From Exact Dynamics to a Numerical Method

Exact Hamiltonian trajectories are rarely available. HMC therefore uses a reversible, volume-preserving numerical integrator and corrects its energy error with a Metropolis step. The usual kinetic energy is $K(p) = \|p\|_2^2/2$, or more generally $K(p) = p^\top M^{-1}p/2$ with mass matrix M chosen to match posterior scale.

3.10.2 Key Properties

Proposition 3.10.1. *Because the leapfrog proposal is reversible up to momentum flip and preserves volume, accepting it with probability* $1 \wedge \exp[-H(q',p') + H(q,p)]$ *leaves the joint target* $e^{-H(q,p)}$ *invariant.*

3.10.3 Leapfrog Integrator

Lemma 3.10.1. *The leapfrog integrator preserves volume and is reversible up to momentum flip. It is a second-order integrator: over a fixed integration interval, its global state error is* $O(h^2)$ *under standard smoothness assumptions.*

Remark 3.10.1. It does *not* exactly preserve the Hamiltonian.

Example 3.10.1 (Gaussian target)**.** For $U(q) = q^2/2$ and $K(p) = p^2/2$, Hamilton's equations give $\dot{q} = p$, $\dot{p} = -q$, hence

$$\begin{pmatrix} q(t) \\ p(t) \end{pmatrix} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \begin{pmatrix} q(0) \\ p(0) \end{pmatrix}.$$

The trajectory is a circle of constant energy. Integration time near a multiple of 2π returns close to the starting point, explaining why HMC randomizes trajectory length or uses the no-U-turn criterion.

HMC with Leapfrog

Algorithm 10 Hamiltonian Monte Carlo (practical)

Sample $p \sim e^{-K(p)}$
Simulate Hamiltonian dynamics via leapfrog for time T
Accept with probability

$$\alpha = \min\{e^{-H(q',p')+H(q,p)}, 1\}$$

Remark 3.10.2. The Metropolis step corrects discretization error.

Choosing Parameters

- Integration time T:
  - Too small ⇒ slow exploration
  - Too large ⇒ wasted computation
- Randomizing T helps avoid periodic behavior
- No-U-Turn criterion adaptively selects trajectory length

Geometric Adaptation

Definition 3.10.1. Riemannian HMC uses the position-dependent kinetic energy

$$K(q,p) = \frac{1}{2}p^\top \Sigma^{-1}(q)p + \frac{1}{2}\log\det\Sigma(q).$$

3.11 Convergence of Markov Chains

3.11.1 Spectral Gap

Theorem 3.11.1 (Reversible spectral contraction)**.** *Let K be reversible with respect to μ, and let $1 = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_m \geq -1$ be its eigenvalues on a finite state space. Define $\lambda_* = \max_{j\geq 2}|\lambda_j|$. Then*

$$\mathrm{Var}_\mu(K^n f) \leq \lambda_*^{2n}\mathrm{Var}_\mu(f).$$

Proof. Reversibility makes K self-adjoint in $L^2(\mu)$. Expand the centered function $f - \mathbb{E}_\mu f$ in an orthonormal eigenbasis excluding the constant eigenfunction. Applying K^n multiplies its jth coefficient by λ_j^n; Parseval's identity then gives the bound. □

The absolute spectral gap is $\gamma_* = 1 - \lambda_*$. A lazy chain has nonnegative eigenvalues, in which case $\lambda_* = \lambda_2$ and the usual gap is $1 - \lambda_2$. Lazification prevents a period-two oscillation caused by an eigenvalue near -1.

3.11.2 Non-reversible Chains

Remark 3.11.1. For non-reversible chains: $\chi^2(\nu K^n||\mu) \leq C|\lambda_2|^{2n}\chi^2(\nu||\mu)$.

Remark 3.11.2. In continuous time this is called *hypocoercivity*.

3.11.2.1 Poincaré Inequality

Definition 3.11.1. A continuous-time reversible generator $\mathcal{L}$ satisfies a Poincaré inequality with constant $c > 0$ if

$$-\langle g, \mathcal{L}g\rangle_\mu \geq c\mathrm{Var}_\mu(g)$$

for every g in its domain.

The minus sign is essential because a Markov generator is negative semidefinite. Differentiating the variance along the semigroup gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Var}_\mu(P_t g) = 2\langle P_t g, \mathcal{L} P_t g\rangle_\mu \le -2c\mathrm{Var}_\mu(P_t g),$$

and Grönwall's inequality yields exponential L^2 convergence at rate $2c$.

Remark 3.11.3. Implies exponential convergence rate e^{-2ct}.

3.11.3 Contraction in Divergences

Definition 3.11.2. A kernel K is contractive if $D_f(\nu K||\mu) \le (1-\kappa) D_f(\nu||\mu)$.

Remark 3.11.4. Implies geometric convergence.

3.11.4 Mixing Time

Definition 3.11.3.

$$t_{\mathrm{mix}}(\varepsilon) = \min\{t : \sup_{\mu_0} \mathrm{TV}(\mu_0 K^t, \mu) \le \varepsilon\}.$$

Definition 3.11.4. Relaxation time: $t_{\mathrm{rel}} = \frac{1}{1-|\lambda_2|}$.

For a finite reversible lazy chain with stationary minimum $\pi_{\min} = \min_x \pi(x)$, spectral contraction gives the representative bound

$$t_{\mathrm{mix}}(\varepsilon) \le t_{\mathrm{rel}} \log \frac{1}{\varepsilon \pi_{\min}}.$$

The dependence on $\pi_{\min}$ reflects worst-case initialization. A warm start replaces it by a smaller density-ratio factor.

Example 3.11.1 (Random walk on a cycle)**.** For the lazy nearest-neighbor walk on a cycle of length n, the spectral gap is of order n^{-2}. Hence relaxation and mixing occur on the diffusive time scale n^2 rather than the ballistic scale n.

3.11.5 Conductance

Definition 3.11.5. For $S \subset \Omega$:

$$\Phi(S) = \frac{Q(S, S^c)}{\mu(S)}, \quad Q(S, S^c) = \sum_{x \in S, y \in S^c} \mu(x) P(x, y).$$

Definition 3.11.6. $\Phi_* = \min_{\mu(S) \le 1/2} \Phi(S)$.

Theorem 3.11.2. $t_{\mathrm{mix}}(1/4) \ge \frac{1}{4\Phi_*}$.

The conductance of a set is the stationary probability flow leaving it, normalized by its mass. A low-conductance set is a bottleneck. Cheeger's inequalities relate geometry and spectrum for reversible chains:

$$\frac{\Phi_*^2}{2} \le 1 - \lambda_2 \le 2\Phi_*.$$

Thus a bottleneck forces a small spectral gap, while a uniform lower bound on conductance gives quantitative mixing.

Example 3.11.2 (Two well-separated modes)**.** If a target places half its mass in each of two regions separated by a high-energy barrier and a local proposal rarely crosses the barrier, choose S as one mode. Then $Q(S, S^c)$ is exponentially small in the barrier height, so conductance predicts exponentially slow mixing even if the chain moves rapidly within each mode.

3.11.6 Asymptotic Variance

Definition 3.11.7. $\sigma_f^2 = \mathrm{Var}_\mu(f(X_0)) + 2\sum_{k=1}^{\infty} \mathrm{Cov}_\mu(f(X_0), f(X_k))$.

Theorem 3.11.3 (Markov-chain central limit theorem)**.** *Under stationarity and suitable geometric ergodicity and moment conditions,*

$$\sqrt{N}\left(\frac{1}{N}\sum_{k=0}^{N-1} f(X_k) - \mathbb{E}_\mu f\right) \xrightarrow{d} \mathcal{N}(0, \sigma_f^2).$$

In particular, $\mathrm{Var}(N^{-1}\sum_{k=0}^{N-1} f(X_k)) \sim \sigma_f^2/N$.

Writing $\rho_k = \mathrm{Corr}(f(X_0), f(X_k))$ gives

$$\tau_{\mathrm{int}} = 1 + 2\sum_{k\ge 1} \rho_k, \qquad \sigma_f^2 = \mathrm{Var}_\mu(f)\tau_{\mathrm{int}}.$$

The effective sample size $N_{\mathrm{eff}} = N/\tau_{\mathrm{int}}$ is observable-specific.

Example 3.11.3 (AR(1) autocorrelation)**.** For $X_{t+1} = \rho X_t + \sqrt{1-\rho^2}Z_t$ in stationarity and $f(x) = x$, $\rho_k = \rho^k$, so

$$\tau_{\mathrm{int}} = \frac{1+\rho}{1-\rho}.$$

At $\rho = 0.9$, one thousand iterations contain only about 53 independent equivalents for estimating the mean.

3.11.7 Practical Diagnostics

Trace plots can reveal drift, sticking, or unexplored modes, but visual stationarity is not a proof of convergence. Autocorrelation and effective sample size quantify Monte Carlo error for selected observables. Multiple overdispersed chains permit between-chain and within-chain comparisons such as $\widehat{R}$; rank normalization improves robustness for heavy-tailed output. Diagnostics should be reported for scientifically relevant functionals and tail probabilities, not only for raw coordinates.

Discussion

A single long chain pays warm-up cost once and may estimate local autocorrelations efficiently. Multiple dispersed chains are better at exposing nonconvergence and mode trapping. Discarding a fixed "burn-in" does not cure a chain whose mixing time exceeds the run length; initialization, geometry, transition design, and convergence diagnostics must be considered together.

3.12 Tempering and Particle Methods

3.12.1 Multimodality and Slow Mixing

In a multimodal target, probability mass concentrates in separated regions. A local random-walk or Langevin proposal must cross low-density energy barriers to move between them, so rapid exploration within each mode can coexist with extremely slow global mixing.

Remark 3.12.1. For distributions of the form $p(x) \propto e^{-f(x)}$, the time to transition between modes is often *exponential* in the barrier height: mixing time $\sim \exp(\Delta f)$. This is known as *metastability*.

3.12.2 Temperature Scaling

Introduce a *temperature parameter* $T > 0$: $p_T(x) \propto e^{-f(x)/T} = e^{-\beta f(x)}, \quad \beta = \frac{1}{T}$.

- High temperature ($T \gg 1$, $\beta \ll 1$): distribution is *flattened*.
- Low temperature ($T \ll 1$, $\beta \gg 1$): distribution concentrates near modes.

Remark 3.12.2. In Langevin dynamics: $dX_t = -\nabla V(X_t)\,dt + \sqrt{2T}\,dB_t$, temperature controls the tradeoff between gradient (drift) and noise.

3.12.3 Simulated Annealing

Simulated annealing gradually lowers temperature: $T_1 > T_2 > \cdots \to 0$.

- Initially explores broadly.
- Eventually concentrates near global minima.

Remark 3.12.3. Once modes separate at low temperature, the chain becomes *trapped*: probability mass cannot easily move between modes.

3.12.4 Simulated Tempering

Introduce temperature as an auxiliary variable: $(\ell, x) \in \{1, \ldots, L\} \times \Omega$.

Define target: $\pi(\ell, x) \propto w_\ell\, p_\ell(x), \quad p_\ell(x) \propto e^{-f(x)/T_\ell}$.

Algorithm:

- Alternate:
  - Update x using MCMC at fixed temperature ℓ.
  - Propose $\ell \to \ell \pm 1$ with MH acceptance:

$$\alpha = \min\left\{\frac{w_{\ell'} p_{\ell'}(x)}{w_\ell p_\ell(x)}, 1\right\}.$$

Remark 3.12.4. Drawback: requires knowledge (or estimation) of normalizing constants for p_ℓ.

3.12.5 Parallel Tempering (Replica Exchange)

Run multiple chains at different temperatures: $(x_1, \ldots, x_L), \quad x_\ell \sim p_\ell$.

Target: $\pi(x_1, \ldots, x_L) = \prod_{\ell=1}^{L} p_\ell(x_\ell)$.

Swap proposal: $(x_i, x_{i+1}) \to (x_{i+1}, x_i)$, with acceptance probability

$$\alpha = \min\left\{\frac{p_i(x_{i+1}) p_{i+1}(x_i)}{p_i(x_i) p_{i+1}(x_{i+1})}, 1\right\}.$$

Remark 3.12.5. Advantages:

- No need for normalizing constants.
- High-temperature chains help low-temperature chains escape modes.

3.12.6 Sequential Monte Carlo (SMC)

Construct a sequence of distributions: $p_k(x) \propto \tilde{p}_k(x), \quad k = 1, \ldots, L$, typically interpolating between easy and target distributions.

Algorithm 11 Sequential Monte Carlo

1: Sample $X_1^{(i)} \sim p_1$
2: **for** $k = 2$ to L **do**
3: Compute weights:

$$w_k^{(i)} \propto \frac{\tilde{p}_k(X_{k-1}^{(i)})}{\tilde{p}_{k-1}(X_{k-1}^{(i)})}$$

4: Resample particles according to $w_k^{(i)}$
5: Move particles: $X_k^{(i)} \sim K_k(X_{k-1}^{(i)}, \cdot)$
6: **end for**
7: Output $\{X_L^{(i)}\}$

Remark 3.12.6. SMC combines importance weighting, resampling, and invariant mutation kernels. The effective sample size

$$\mathrm{ESS}_k = \frac{(\sum_i w_k^{(i)})^2}{\sum_i (w_k^{(i)})^2}$$

diagnoses weight concentration. Resampling prevents a few particles from carrying all mass, but it adds Monte Carlo variance and duplicates ancestral lineages. Mutation kernels restore diversity without changing the current target.

Example 3.12.1 (Bootstrap particle filter)**.** For a state-space model $X_k \sim p(x_k \mid x_{k-1})$ and $Y_k \sim p(y_k \mid x_k)$, propagate through the state transition and weight by $p(y_k \mid X_k^{(i)})$. The particle cloud approximates $p(x_k \mid y_{1:k})$. Highly informative observations can cause weight collapse because the transition prior rarely proposes states compatible with the new data.

3.12.7 Annealed Importance Sampling (AIS)

A simplified version of SMC without resampling.

Algorithm 12 Annealed Importance Sampling

1: Sample $X_1^{(i)} \sim p_1$
2: **for** $k = 2$ to L **do**
3: Update the weight using $X_{k-1}^{(i)}$
4: Sample $X_k^{(i)} \sim K_k(X_{k-1}^{(i)}, \cdot)$
5: **end for**
6: Compute weights:

$$\tilde{w}^{(i)} = \prod_{k=2}^{L} \frac{\tilde{p}_k(X_{k-1}^{(i)})}{\tilde{p}_{k-1}(X_{k-1}^{(i)})}$$

7: Estimate:

$$\mathbb{E}_{p_L}[f(X)] \approx \frac{\sum_i \tilde{w}^{(i)} f(X_L^{(i)})}{\sum_i \tilde{w}^{(i)}}$$

Remark 3.12.7. If p_1 has known normalizer Z_1 and each K_k leaves p_k invariant, then the AIS weight satisfies

$$\mathbb{E}\tilde{w} = \frac{Z_L}{Z_1}.$$

Thus averaging independent weights estimates a partition-function ratio. The ratio estimate is unbiased, although its logarithm is not. A schedule with poor overlap between adjacent targets produces highly variable weights.

Exploration and Concentration

Tempering and particle methods address the same conflict. High-temperature distributions are diffuse and easy to explore but are poor approximations to the target. Low-temperature

distributions concentrate correctly but can be multimodal and difficult to traverse. Tempering moves a chain between these regimes; SMC propagates a population through them; AIS accumulates the corresponding likelihood ratios without resampling.

3.13 Variational Methods

3.13.1 MCMC vs Variational Inference

Two paradigms for approximate inference:

- **MCMC:**
  - Constructs a Markov chain with stationary distribution p
  - Asymptotically exact
  - Slow mixing; hard to diagnose convergence
- **Variational methods:**
  - Solve an optimization problem over distributions
  - Fast and scalable
  - Approximate; may converge to local optima

Remark 3.13.1. MCMC: accuracy ↑, speed ↓
Variational inference: speed ↑, accuracy ↓ (approximation bias)

3.13.2 Gibbs Variational Principle

Let

$$p(x) = \frac{e^{f(x)}}{Z}, \quad Z = \int e^{f(x)} dx.$$

Lemma 3.13.1 (Gibbs variational principle)**.**

$$\ln Z = \max_q \left\{ H(q) + \mathbb{E}_{x\sim q}[f(x)] \right\}.$$

The maximum is attained at $q = p$.

Proof. Compute:

$$\mathrm{KL}(q\|p) = \int q(x) \log \frac{q(x)}{p(x)} dx = \int q(x) \log q(x) dx - \int q(x) f(x) dx + \log Z.$$

Rearrange: $\log Z = H(q) + \mathbb{E}_q[f(x)] + \mathrm{KL}(q\|p)$.

Since $\mathrm{KL}(q\|p) \geq 0$, $\log Z \geq H(q) + \mathbb{E}_q[f(x)]$, with equality iff $q = p$. □

3.13.3 Variational Relaxation

We cannot optimize over all distributions q, so we restrict the search to a class $\mathcal{Q}$:

$$\log Z \geq \max_{q \in \mathcal{Q}} \{H(q) + \mathbb{E}_q[f(x)]\}.$$

Remark 3.13.2. Tradeoff:

- Larger $\mathcal{Q}$: better approximation, harder optimization
- Smaller $\mathcal{Q}$: easier optimization, worse approximation

3.13.4 KL Projection Interpretation

Using the identity: $\log Z = \mathrm{KL}(q\|p) + H(q) + \mathbb{E}_q[f(x)]$,

we obtain:

$$\arg\min_{q \in \mathcal{Q}} \mathrm{KL}(q\|p) = \arg\max_{q \in \mathcal{Q}} \{H(q) + \mathbb{E}_q[f(x)]\}.$$

Remark 3.13.3. Variational inference = **projection of p onto $\mathcal{Q}$ in KL divergence**.

3.13.5 Two Use Cases

Partition function approximation:

$$\log Z \geq \max_{q \in \mathcal{Q}} \{H(q) + \mathbb{E}_q[f(x)]\}.$$

Distribution approximation:

$$q^* = \arg\min_{q \in \mathcal{Q}} \mathrm{KL}(q\|p).$$

For latent-variable models: $p(x) = \int p(x,z)dz$, we obtain: $\log p(x) = \mathrm{KL}(q(z)\|p(z|x)) + H(q) + \mathbb{E}_q[\log p(x,z)]$.

3.13.6 Evidence Lower Bound (ELBO)

Definition 3.13.1. $\mathrm{ELBO}(q) = H(q) + \mathbb{E}_{z\sim q}[\log p(x,z)]$.

$$\log p(x) \geq \mathrm{ELBO}(q).$$

Remark 3.13.4. Maximizing ELBO is equivalent to minimizing: $\mathrm{KL}(q(z)\|p(z|x))$.

3.13.7 Mean-Field Variational Inference

Assume factorization: $q(x_1, \ldots, x_n) = \prod_{i=1}^{n} q_i(x_i)$.

Optimize coordinate-wise: $q_i^*(x_i) \propto \exp\left(\mathbb{E}_{q_{-i}}[f(x)]\right)$.

To derive the update, hold q_{-i} fixed and write the ELBO terms depending on q_i as

$$\int q_i(x_i)\mathbb{E}_{q_{-i}}[\log p(x_i, X_{-i})]\,\mathrm{d}x_i + H(q_i).$$

Adding a Lagrange multiplier for $\int q_i = 1$ and taking a functional derivative gives

$$\log q_i^*(x_i) = \mathbb{E}_{q_{-i}}[\log p(x_i, X_{-i})] + \text{constant}.$$

Each exact coordinate update cannot decrease the ELBO. The algorithm converges to a coordinatewise optimum, but the nonconvex mean-field problem may have multiple local optima.

Algorithm 13 Coordinate Ascent Variational Inference (CAVI)

1: Initialize $q_1, \ldots, q_n$
2: **repeat**
3: **for** $i = 1, \ldots, n$ **do**
4: $$q_i(x_i) \propto \exp\left(\mathbb{E}_{q_{-i}}[f(x)]\right)$$
5: **end for**
6: **until** convergence

Example 3.13.1 (Mean field for the Ising model)**.** For $p(x) \propto \exp(\sum_{i<j} J_{ij}x_ix_j + \sum_i h_ix_i)$ and $q(x) = \prod_i q_i(x_i)$, let $m_i = \mathbb{E}_q X_i$. The coordinate update gives

$$q_i(x_i) \propto \exp\left[x_i\left(h_i + \sum_{j\neq i} J_{ij}m_j\right)\right],$$

or equivalently

$$m_i = \tanh\left(h_i + \sum_{j\neq i} J_{ij}m_j\right).$$

These fixed-point equations can have multiple solutions in strongly coupled models, reflecting both physical phase transitions and nonconvex variational optimization.

3.13.8 Approximate Likelihood-Based Learning

For latent-variable models $p_\theta(x, z)$: $\log p_\theta(x) = \max_q \{H(q) + \mathbb{E}_q[\log p_\theta(x, z)]\}$.

Instead optimize:

$$\max_\theta \max_q \sum_{i=1}^{N} \left[H(q_i) + \mathbb{E}_{z\sim q_i} \log p_\theta(x_i, z)\right].$$

If the variational family excludes the true posterior, the optimized ELBO differs from the log likelihood by the approximation gap $\min_{q\in\mathcal{Q}} \mathrm{KL}(q(z)\|p_\theta(z \mid x))$. When the same restricted family is used while learning θ, this gap depends on θ and can bias parameter estimates toward models with easier posteriors.

3.13.9 Amortized Variational Inference

Parameterize: $q_\phi(z|x)$.

Objective:

$$\max_{\theta,\phi} \sum_{i=1}^{N} \mathbb{E}_{z\sim q_\phi(z|x_i)} \left[\log \frac{p_\theta(x_i, z)}{q_\phi(z|x_i)}\right].$$

Amortization replaces one optimization problem per observation by one learned map $x \mapsto q_\phi(\cdot \mid x)$. The total inference error can be decomposed into a variational-family gap and an *amortization gap*: even the best encoder parameters may not attain the separately optimized variational distribution for every x. Iterative refinement at test time can reduce the latter without changing the family.

3.13.10 Variational Autoencoder (VAE)

A VAE specifies a latent prior $p(z)$, a decoder $p_\theta(x \mid z)$, and an amortized encoder $q_\phi(z \mid x)$. Inserting the encoder into the marginal likelihood gives the exact identity

$$\begin{aligned}\log p_\theta(x) &= \mathbb{E}_{q_\phi(z|x)} \log \frac{p_\theta(x, z)}{q_\phi(z \mid x)} + \mathrm{KL}\Big(q_\phi(z \mid x)\|p_\theta(z \mid x)\Big) \\ &= \underbrace{\mathbb{E}_{q_\phi} \log p_\theta(x \mid z) - \mathrm{KL}(q_\phi(z \mid x)\|p(z))}_{\mathcal{L}_{\mathrm{VAE}}(x)} + \mathrm{KL}(q_\phi(z \mid x)\|p_\theta(z \mid x)).\end{aligned}$$

The ELBO is tight precisely when the encoder equals the model posterior. The first term rewards accurate conditional reconstruction, while the KL term makes the aggregate latent representation compatible with sampling from the prior. Their competition is a rate–distortion tradeoff, not merely an informal reconstruction-versus-regularization heuristic.

Example 3.13.2 (Gaussian encoder)**.** Take a standard Gaussian prior and a diagonal Gaussian encoder:

$$p(z) = \mathcal{N}(0, I), \qquad q_\phi(z \mid x) = \mathcal{N}\Big(\mu_\phi(x), \mathrm{diag}(\sigma_\phi(x)^2)\Big).$$

Then

$$\mathrm{KL}(q_\phi\|p) = \frac{1}{2}\sum_j \left(\mu_j(x)^2 + \sigma_j(x)^2 - 1 - \log \sigma_j(x)^2\right),$$

an analytic penalty. Samples are reparameterized as $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, allowing low-variance pathwise gradients.

If the decoder is too expressive, it may model x without using z, causing *posterior collapse*: $q_\phi(z \mid x) \approx p(z)$. KL warm-up, free-bits objectives, architectural restrictions, and richer priors are common responses, but each changes the rate–distortion balance.

Example 3.13.3 (Linear Gaussian VAE)**.** Let $Z \sim \mathcal{N}(0, I_k)$ and $X \mid Z = z \sim \mathcal{N}(Wz, \sigma^2 I_d)$. Then

$$X \sim \mathcal{N}(0, WW^\top + \sigma^2 I_d),$$

and Gaussian conditioning gives

$$Z \mid X = x \sim \mathcal{N}(Mx, C), \quad C = (I + \sigma^{-2} W^\top W)^{-1}, \quad M = \sigma^{-2} C W^\top.$$

An affine Gaussian encoder can make the ELBO exact. A diagonal-covariance encoder incurs a variational gap whenever C has important off-diagonal entries. This solvable model isolates approximation error from neural-network optimization error.

3.13.11 Gradient Estimators

Score-function (REINFORCE):

$$\nabla_\phi \mathbb{E}_{z\sim q_\phi}[f(z)] = \mathbb{E}_{z\sim q_\phi}\left[f(z) \nabla_\phi \log q_\phi(z)\right].$$

Reparameterization trick:

$$z = g(x, \xi, \phi), \quad \xi \sim p(\xi),$$

$$\nabla_\phi \mathbb{E}_{z\sim q_\phi}[f(z)] = \mathbb{E}_\xi[\nabla_\phi f(g(x, \xi, \phi))].$$

Example 3.13.4.

$$z \sim \mathcal{N}(\mu_\phi(x), \Sigma_\phi(x)) \quad \Rightarrow \quad z = \mu_\phi(x) + \Sigma_\phi^{1/2}(x)\, \xi.$$

Remark 3.13.5.

- REINFORCE: general but high variance
- Reparameterization: lower variance but requires differentiability

3.13.12 Expectation Maximization (EM)

We consider latent-variable models $p_\theta(x, z)$ with observed data $x_1, \ldots, x_n$.

$$\max_\theta \sum_{i=1}^n \log p_\theta(x_i) = \max_\theta \sum_{i=1}^n \log \int p_\theta(x_i, z) dz.$$

Using the variational bound,

$$\log p_\theta(x_i) \geq \max_{q_i} \left[H(q_i) + \mathbb{E}_{z\sim q_i} \log p_\theta(x_i, z)\right].$$

Algorithm 14 Expectation Maximization (EM)

1: Initialize θ
2: **repeat**
3: **E-step:**

$$q_i(z) \leftarrow p_\theta(z|x_i)$$

4: **M-step:**

$$\theta \leftarrow \arg\max_\theta \sum_{i=1}^{n} \mathbb{E}_{z\sim q_i}[\log p_\theta(x_i, z)]$$

5: **until** convergence

Remark 3.13.6. EM alternates between:

- computing posteriors over latent variables
- maximizing expected complete-data log-likelihood

3.13.13 Example: Mixture of Gaussians

Model: $Z_m \sim \text{Categorical}(\pi), \quad X_m \sim \mathcal{N}(\mu_{Z_m}, \Sigma_{Z_m})$.

E-step: posterior responsibilities

$$z_{mj} := p(Z_m = j|X_m = x_m) = \frac{\pi_j \mathcal{N}(x_m; \mu_j, \Sigma_j)}{\sum_k \pi_k \mathcal{N}(x_m; \mu_k, \Sigma_k)}.$$

M-step:

$$\pi_j \leftarrow \frac{1}{n}\sum_{m=1}^{n} z_{mj},$$

$$\mu_j \leftarrow \frac{\sum_{m=1}^{n} z_{mj} x_m}{\sum_{m=1}^{n} z_{mj}},$$

$$\Sigma_j \leftarrow \frac{\sum_{m=1}^{n} z_{mj}(x_m - \mu_j)(x_m - \mu_j)^\top}{\sum_{m=1}^{n} z_{mj}}.$$

3.13.14 Outer Relaxation

Instead of optimizing over global distributions $q(x)$, optimize over local marginals.

Let $p(x) \propto \exp\left(\sum_C \phi_C(x_C)\right)$.

Define pseudo-marginals $\{q_C\}$: $\mathbb{E}_q\left[\sum_C \phi_C(x_C)\right] = \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)]$.

Relaxation:

$$\log Z \approx \max_{\{q_C\}\in\mathcal{Q}} \left[H(q) + \sum_C \mathbb{E}_{q_C}[\phi_C(x_C)]\right].$$

Remark 3.13.7. Constraints:

- consistency between marginals
- $q_C(x_C) \geq 0$, $\sum q_C = 1$

3.13.15 Marginal Polytope vs Local Polytope

- **Marginal polytope:** exact set of realizable marginals (intractable)
- **Local polytope:** enforce only pairwise consistency

Remark 3.13.8. Membership in marginal polytope is NP-hard.

3.13.16 Tree Case (Exact)

Lemma 3.13.2. *If the factor graph is a tree, local consistency implies global consistency.*

$$q(x) = \frac{\prod_{a \in F} q_a(x_a)}{\prod_{i \in V} q_i(x_i)^{d_i - 1}}.$$

3.13.17 Bethe Entropy

Lemma 3.13.3. *For tree-structured graphs,* $H(q) = \sum_{a \in F} H(q_a) - \sum_{i \in V} (d_i - 1) H(q_i)$.

Definition 3.13.2. The RHS defines the **Bethe entropy**: $H_{\text{Bethe}}(\{q_a\}) := \sum_a H(q_a) - \sum_i (d_i - 1) H(q_i)$.

3.13.18 Bethe Free Energy

Optimization problem:

$$\max_{\{q_a\} \in \mathcal{Q}} \left[H_{\text{Bethe}}(\{q_a\}) + \sum_a \mathbb{E}_{q_a}[\phi_a(x_a)] \right].$$

Remark 3.13.9.

- Exact on trees
- Approximate on loopy graphs
- Not necessarily convex

Note. Stationary points correspond to fixed points of loopy belief propagation.

3.13.19 KL Projections

Two types:

- **I-projection:**
$$\min_{q \in \mathcal{Q}} \mathrm{KL}(q \| p)$$
underestimates variance
- **M-projection:**
$$\min_{q \in \mathcal{Q}} \mathrm{KL}(p \| q)$$
matches moments

3.13.20 M-Projection for Exponential Families

Let
$$q_\theta(x) = \frac{\exp(\langle \theta, \phi(x) \rangle)}{Z(\theta)}.$$

Lemma 3.13.4 (Moment matching)**.**
$$\arg\min_\theta \mathrm{KL}(p \| q_\theta) = \theta^* \quad s.t. \quad \mathbb{E}_{q_{\theta^*}}[\phi(x)] = \mathbb{E}_p[\phi(x)].$$

3.13.21 Belief Propagation as Variational Inference

Messages correspond to approximate marginals: $\nu_{i \to a}(x_i) \approx$ marginal of x_i.

Updates correspond to projections onto exponential family approximations.

Remark 3.13.10. Variational view:

- BP = coordinate ascent on Bethe free energy
- Loopy BP = approximate inference via Bethe relaxation

3.13.22 Example: Bayesian Linear Model

$$y = Ax + w, \quad w \sim \mathcal{N}(0, \sigma^2 I).$$

Posterior: $p(x|y) \propto \exp\left(-\frac{1}{2\sigma^2}\|y - Ax\|^2\right) p_0(x)$.

If $p_0(x) \propto e^{-\beta\|x\|_1}$, then MAP becomes: $\min_x \|y - Ax\|^2 + \lambda\|x\|_1$, i.e., LASSO.

3.13.23 Approximate Message Passing (AMP)

To obtain tractable updates:

- Assume large random matrix A
- Use Gaussian approximations for messages

Remark 3.13.11. AMP emerges as a simplification of belief propagation under high-dimensional limits.

Chapter 4

Modern Deep Generative Models

4.1 Overview of Deep Generative Modeling

The goal is to learn a distribution p_θ from samples $x_1, \dots, x_N \sim p_{\text{data}}$ and to generate new, nonmemorized samples. A useful taxonomy asks which operations are tractable.

Normalizing flows provide exact density evaluation and direct sampling by restricting the generator to an invertible map. VAEs permit noninvertible latent-variable decoders but replace exact likelihood by a variational lower bound. Energy-based models specify an unnormalized density and therefore need MCMC, variational inference, or density-ratio methods. GANs avoid likelihoods and learn through a discriminator. Diffusion and flow-matching models learn local dynamics connecting noise and data, then solve an SDE or ODE to sample.

No family dominates all criteria. Exact likelihood can constrain architecture; latent variables create an inference problem; adversarial objectives can be unstable; and dynamic models may require many network evaluations. The relevant comparison includes sample quality, mode coverage, likelihood or calibration, training stability, and sampling cost.

4.2 Generative Modeling via Transformations

Let $z \sim p_0(z)$ be a simple distribution (e.g., $\mathcal{N}(0, I)$), and define

$$x = g_\theta(z).$$

Then p_θ is the pushforward: $p_\theta = (g_\theta)_{\#} p_0$.

If g_θ is invertible: $p_\theta(x) = p_0(g_\theta^{-1}(x)) \left| \det Dg_\theta^{-1}(x) \right|$.

Remark 4.2.1. This is the foundation of **normalizing flows**.

4.3 Normalizing Flows

4.3.1 Definition

Let $z \sim p_0(z)$ and $x = g(z)$ with invertible g. Then: $\log p(x) = \log p_0(z) + \log |\det Dg^{-1}(x)|$.

Remark 4.3.1. Key requirement: efficient computation of $\det Dg^{-1}(x)$.

Example 4.3.1 (One-dimensional monotone flow)**.** Let $Z \sim \mathcal{N}(0, 1)$ and let g be strictly increasing. Then

$$p_X(x) = \frac{\varphi(g^{-1}(x))}{g'(g^{-1}(x))}.$$

Choosing $g = F_X^{-1} \circ \Phi$ transports a standard Gaussian exactly to any continuous target CDF F_X. Multivariate flows generalize this idea, but there is no single canonical ordering of multivariate quantiles.

4.3.2 Composition of Flows

Let

$$g = g_L \circ \cdots \circ g_1.$$

Then: $\log |\det Dg^{-1}(x)| = \sum_{\ell=1}^{L} \log |\det Df_\ell(x_\ell)|$, where $f_\ell = g_\ell^{-1}$.

Remark 4.3.2. Design principle: compose simple invertible transformations with tractable Jacobians.

4.4 Building Blocks for Flows

4.4.1 Affine Transformations

$$f(x) = Ax + b.$$

- $\det Df = \det A$
- Efficient if A is triangular
- Limited expressivity

4.4.2 Coordinate-wise Transformations

$$f(x) = (f_1(x_1), \ldots, f_d(x_d)).$$

$$\det Df = \prod_{i=1}^{d} f_i'(x_i).$$

Remark 4.4.1. Keeps coordinates independent $\Rightarrow$ limited modeling power.

4.5 Affine Coupling Layers

Partition variables: $x = (x_1, x_2)$.

Define: $z_1 = x_1, \quad z_2 = x_2 \odot \exp(s(x_1)) + t(x_1)$.

Jacobian is triangular: $\log|\det Df(x)| = \sum_i s_i(x_1)$.

The inverse is explicit:

$$x_1 = z_1, \qquad x_2 = (z_2 - t(z_1)) \odot e^{-s(z_1)}.$$

Although s and t may be arbitrarily complicated neural networks, they do not need to be invertible because the unchanged block x_1 is available when inverting the layer. Alternating partitions or inserting invertible channel mixing lets every coordinate eventually influence every other coordinate.

Remark 4.5.1.

- Efficient determinant: $\mathcal{O}(d)$
- Used in NICE, RealNVP, Glow

4.6 Residual Flows

Lemma 4.6.1. *Let $f(x) = x + F(x)$ where F is L-Lipschitz with $L < 1$. Then f is bijective.*

Proof. For a desired output y, solving $f(x) = y$ is equivalent to finding a fixed point of $T_y(x) = y - F(x)$. Since T_y is an L-contraction, Banach's theorem gives a unique solution. Moreover, $\|f(x) - f(x')\| \geq (1 - L)\|x - x'\|$, so the inverse is Lipschitz. □

Theorem 4.6.1 (Banach fixed point)**.** *If T is a contraction, then it has a unique fixed point.*

Remark 4.6.1. Residual flows ensure invertibility via contraction mappings.

4.7 Efficient Log-Determinant Computation

$$\log\det(I + DF) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \operatorname{Tr}((DF)^k).$$

4.7.1 Hutchinson Trace Estimator

Lemma 4.7.1. *For $v \sim \mathcal{N}(0, I)$:* $\operatorname{Tr}(A) = \mathbb{E}[v^\top A v]$.

Remark 4.7.1. Allows stochastic estimation of log-determinants in high dimensions.

4.8 Flows in Variational Inference

Let

$$z = f_\phi(u, x), \quad u \sim \mathcal{N}(0, I).$$

Then ELBO becomes: $\mathbb{E}_u \left[\log p_\theta(x, f_\phi(u, x)) - \log q_\phi(f_\phi(u, x)|x)\right]$.

Remark 4.8.1. Flows increase expressivity of variational distributions.

4.9 Pros and Cons of Normalizing Flows

Pros:

- Exact likelihood evaluation
- Efficient sampling
- Invertible mapping enables interpolation

Cons:

- Requires invertibility constraints
- Jacobian computation can be costly
- May struggle with topological mismatch

4.10 Continuous-Time Flows (Neural ODEs)

Define dynamics: $\frac{dx(t)}{dt} = F(x(t), t), \quad x(0) = x_0$.

Theorem 4.10.1 (Picard–Lindelöf)**.** *If F is Lipschitz, then the ODE has a unique solution.*

Remark 4.10.1. Defines a continuous flow: $x(T) = f_T(x_0)$.

4.11 Density Evolution

Let X_t solve the deterministic ODE $\dot{X}_t = F(X_t, t)$ and suppose its law has a smooth density p_t. Conservation of probability gives the *continuity equation*

$$\partial_t p_t(x) + \nabla \cdot \Big(p_t(x) F(x, t)\Big) = 0.$$

Along a characteristic $t \mapsto X_t$, the chain rule therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \log p_t(X_t) = \partial_t \log p_t(X_t) + F(X_t, t)^\top \nabla \log p_t(X_t) = -\nabla \cdot F(X_t, t).$$

Thus a continuous normalizing flow can evaluate likelihoods by integrating the divergence:

$$\log p_T(X_T) = \log p_0(X_0) - \int_0^T \nabla \cdot F(X_t, t)\ \mathrm{d}t.$$

Remark 4.11.1. The formula uses the divergence evaluated along the trajectory. Writing $\partial_t \log p_t(x) = -\nabla \cdot F(x,t)$ at a fixed spatial point would omit the transport term $F^\top \nabla \log p_t$.

4.12 Score-Based Modeling

4.12.1 Score Function

Definition 4.12.1. The score function is: $s(x) = \nabla \log p(x)$.

4.12.2 Score Matching

For a data density p and a vector field s_θ, the Fisher divergence is

$$J(\theta) = \frac{1}{2}\mathbb{E}_p \left\| s_\theta(X) - \nabla \log p(X) \right\|_2^2.$$

Direct evaluation is impossible because the data score $\nabla \log p$ is unknown. The following identity removes it.

Proposition 4.12.1 (Hyvärinen score-matching identity)**.** *Assume p and s_θ are continuously differentiable, $\mathbb{E}_p \left\| s_\theta(X) \right\|^2 < \infty$, and $p(x) s_{\theta,j}(x) \to 0$ at the boundary of the support for every coordinate j. Then*

$$J(\theta) = \mathbb{E}_p \left[\frac{1}{2} \left\| s_\theta(X) \right\|_2^2 + \nabla \cdot s_\theta(X) \right] + C_p,$$

where $C_p = \frac{1}{2}\mathbb{E}_p \left\| \nabla \log p(X) \right\|_2^2$ is independent of θ.

Proof. Expand the square. The only term involving the unknown score and θ is

$$\mathbb{E}_p \langle s_\theta, \nabla \log p \rangle = \sum_j \int s_{\theta,j}(x)\, \partial_j p(x)\ \mathrm{d}x.$$

Integration by parts and the boundary assumption turn this into $-\mathbb{E}_p[\nabla \cdot s_\theta(X)]$. Substitution into the expansion proves the claim. □

Remark 4.12.1. If $p_\theta(x) = \widetilde{p}_\theta(x)/Z_\theta$, then $\nabla_x \log p_\theta(x) = \nabla_x \log \widetilde{p}_\theta(x)$; the unknown normalizing constant disappears because it does not depend on x.

4.12.3 Integration by Parts Trick

Up to a positive scale and a constant independent of θ, the empirical objective is therefore

$$\widehat{J}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{1}{2}\left\|s_\theta(X_i)\right\|_2^2 + \operatorname{tr}\Big(\nabla_x s_\theta(X_i)\Big)\right].$$

The trace is expensive in high dimension. If V has independent, mean-zero, unit-variance coordinates, Hutchinson's identity gives

$$\operatorname{tr}(\nabla s_\theta(x)) = \mathbb{E}_V[V^\top \nabla s_\theta(x) V],$$

which can be estimated using Jacobian–vector products.

4.13 Denoising Score Matching

Add noise: $y = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 4.13.1 (Tweedie's formula)**.** *Let $Y = X + \sigma Z$, where $Z \sim \mathcal{N}(0, I)$ is independent of X, and let p_σ denote the density of Y. Then*

$$\nabla_y \log p_\sigma(y) = \frac{\mathbb{E}[X \mid Y = y] - y}{\sigma^2}.$$

Proof. Differentiate the Gaussian convolution under the integral sign:

$$\begin{aligned}\nabla p_\sigma(y) &= \int p_X(x)\nabla_y \varphi_{\sigma^2 I}(y - x)\,\mathrm{d}x \\ &= \int p_X(x)\varphi_{\sigma^2 I}(y - x)\frac{x - y}{\sigma^2}\,\mathrm{d}x.\end{aligned}$$

Dividing by $p_\sigma(y)$ converts the normalized integrand into the posterior density of X given $Y = y$. □

Consequently, the denoising objective

$$\mathbb{E}\left\|s_\theta(Y, \sigma) + \frac{Y - X}{\sigma^2}\right\|_2^2$$

has population minimizer $s^*(y, \sigma) = \nabla \log p_\sigma(y)$. Training at many noise scales estimates the scores of an entire path of smoothed data distributions rather than only the often-irregular data density.

Remark 4.13.1. Links score estimation to denoising.

4.14 Diffusion Models (Preview)

For the forward Brownian noising process $\mathrm{d}X_t = \sqrt{2}\,\mathrm{d}B_t$, the density satisfies the heat equation $\partial_t p_t = \Delta p_t$. When the process is parameterized backward from time T to 0, its drift depends on the time-dependent score:

$$\mathrm{d}X_t = -2\nabla \log p_t(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\overline{B}_t, \qquad t : T \downarrow 0.$$

The sign must be interpreted together with the backward time increment $\mathrm{d}t < 0$. Equivalently, after introducing forward reverse-time $\tau = T - t$, the drift points in the denoising direction.

Remark 4.14.1. Learning the score allows sampling from p via reverse-time SDE.

4.15 Summary

The constructions in this chapter differ in what they make tractable. Normalizing flows specify an invertible transport and therefore permit exact density evaluation by change of variables. Neural ODEs replace the discrete composition by a continuous flow and compute density change through the divergence of its velocity field. Score models do not directly normalize a density; they estimate its logarithmic gradient. Diffusion models turn that local object into a sampler by reversing a prescribed noising process. Exact likelihood, efficient sampling, and flexible support impose different computational requirements, and no one family optimizes all three.

Chapter 5

Modern Deep Generative Models (II): Diffusion and Score-Based Methods

5.1 Statistical Complexity of Score Matching

Definition 5.1.1. The **log-Sobolev constant** of a distribution Q on $\mathbb{R}^d$ is the smallest C_{LS} such that, for every distribution P,

$$\mathrm{KL}(P\|Q) \le C_{\mathrm{LS}}\,\mathrm{FI}(P\|Q),$$

where FI is the Fisher divergence.

Remark 5.1.1. The log-Sobolev inequality is stronger than a Poincaré inequality and has a direct dynamical meaning. If P_t evolves under Langevin diffusion with stationary law Q, de Bruijn's identity gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(P_t\|Q) = -\mathrm{FI}(P_t\|Q).$$

Combining this identity with the definition above and applying Grönwall's inequality yields

$$\mathrm{KL}(P_t\|Q) \le e^{-t/C_{\mathrm{LS}}}\mathrm{KL}(P_0\|Q).$$

Thus the same functional inequality that compares a density ratio with its gradient controls the rate at which Langevin dynamics forgets its initialization.

5.1.1 Statistical Efficiency of Score Matching

Theorem 5.1.1 (Forbes & Lauritzen, 2015)**.** *Let $X_1, \dots, X_N \sim P_{\theta^*}$. Then the score matching estimator*

$$\hat{\theta}_{\mathrm{SM}} = \arg\min_{\theta} \sum_{i=1}^{N} \left[\frac{1}{2}\|\nabla \log p_\theta(X_i)\|^2 + \mathrm{Tr}(\nabla^2 \log p_\theta(X_i))\right]$$

satisfies

$$\sqrt{N}(\hat{\theta}_{\mathrm{SM}} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \Gamma_{\mathrm{SM}}).$$

Remark 5.1.2. Score matching is typically less statistically efficient than MLE.

Theorem 5.1.2 (Koehler et al., 2022)**.** *If P_{θ^*} satisfies a restricted Poincaré inequality, then*

$$\|\Gamma_{\mathrm{SM}}\| \lesssim C_P^2 \, \|I(\theta)\|^{-1} \cdot \textit{(moment terms)}.$$

Remark 5.1.3. Statistical gap depends on smoothness and functional inequalities.

5.2 Score Matching on Discrete Spaces

Definition 5.2.1. Let $G = (V, E)$ be a graph. Define the discrete gradient: $\nabla f(v) = (f(w) - f(v))_{w \in N(v)}$.

Example (hypercube $\{\pm 1\}^n$): $\nabla_i f(x) = f(x^{(i)}) - f(x)$, where $x^{(i)}$ flips coordinate i.

Unlike the continuous score, the discrete difference $\nabla_i \log p(x) = \log[p(x^{(i)})/p(x)]$ encodes a local density ratio. These ratios determine a strictly positive distribution up to normalization whenever the state graph is connected: multiply ratios along a path from a fixed reference state to x. Cycle consistency ensures that the result does not depend on the chosen path.

5.3 Pseudo-Likelihood

Definition 5.3.1. The pseudo-likelihood objective: $L_P(q) = \sum_{i=1}^{d} \mathbb{E}_{X \sim p}[\log q(X_i | X_{-i})]$.

The population objective has an exact KL decomposition:

$$L_P(p) - L_P(q) = \sum_{i=1}^{d} \mathbb{E}_{X_{-i} \sim p} \mathrm{KL}\Big(p(\cdot \mid X_{-i}) \| q(\cdot \mid X_{-i})\Big) \geq 0.$$

Thus the true conditionals maximize pseudo-likelihood. If the parametric family is identifiable from its full conditionals and an appropriate uniform law of large numbers holds, the empirical pseudo-likelihood estimator is consistent even though the global partition function is never evaluated.

Remark 5.3.1. The KL decomposition explains both the appeal and the limitation of pseudo-likelihood. It is a proper population criterion and avoids the global normalizing constant, which makes it useful for Ising models and other Markov random fields. It can nevertheless be less statistically efficient than maximum likelihood because each term uses only one conditional view of the joint law.

5.3.1 Generalized Pseudo-Likelihood

Partition coordinates into blocks $S_1, \ldots, S_k$: $L_P(q) = \sum_{i=1}^{k} \mathbb{E}_{X \sim p}[\log q(X_{S_i} | X_{S_i^c})]$.

Larger blocks contain more joint information and can improve statistical efficiency, but their conditional normalizers are more expensive. Ordinary pseudo-likelihood uses singleton blocks; full likelihood is recovered when a block contains every coordinate.

5.4 Diffusion Models

5.4.1 Goal

Diffusion modeling constructs a tractable path of distributions $(p_t)_{0\le t\le T}$ from the data law p_0 to a simple reference law p_T. Training estimates enough local information about this path to reverse it. Generation starts at $X_T \sim p_T$ and numerically transports the sample back to time zero. The central analytic question is therefore not merely how to transform noise into data, but which quantity determines the time reversal of a stochastic process. For diffusions with state-independent noise, that quantity is the score $\nabla \log p_t$.

5.4.2 Forward Process

Let the forward process solve

$$\mathrm{d}X_t = f(X_t, t)\ \mathrm{d}t + G(X_t, t)\ \mathrm{d}W_t, \qquad X_0 \sim p_{\mathrm{data}}.$$

Its coefficients are chosen so that the terminal law p_T is close to a tractable reference distribution such as $\mathcal{N}(0, I)$. If $a(x,t) = G(x,t)G(x,t)^\top$, then the density evolves according to

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2}\sum_{i,j} \partial_{ij}(a_{ij} p_t).$$

This Fokker–Planck equation is the distribution-level counterpart of the forward SDE.

5.4.3 Reverse Process

Theorem 5.4.1 (Anderson, 1982)**.** *Assume p_t is positive and sufficiently smooth. Parameterized with the same time variable running from T down to 0, the reverse-time SDE is*

$$\mathrm{d}X_t = \left[f(X_t, t) - \frac{\nabla \cdot (a p_t)(X_t)}{p_t(X_t)} \right] \mathrm{d}t + G(X_t, t)\ \mathrm{d}\overline{W}_t.$$

Here $[\nabla \cdot (ap)]_i = \sum_j \partial_j (a_{ij} p)$ and $\mathrm{d}t < 0$. If $G(x,t) = g(t)I$, this simplifies to

$$\mathrm{d}X_t = \Big[f(X_t, t) - g(t)^2 \nabla \log p_t(X_t) \Big]\, \mathrm{d}t + g(t)\ \mathrm{d}\overline{W}_t.$$

Remark 5.4.1. Reverse process depends on score $\nabla \log p_t(x)$.

5.4.4 Reverse Brownian Motion

For $\mathrm{d}X_t = \sqrt{2}\ \mathrm{d}W_t$, the reverse drift is $-2\nabla \log p_t$. It is generally *not* ordinary Brownian motion run backward: the score drift is what steers terminal Gaussian noise back toward the data distribution.

5.5 Score-Based Diffusion Framework

Fix a forward SDE

$$\mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t$$

whose transition kernel $q_t(x_t \mid x_0)$ can be sampled. A time-conditioned network $s_\theta(x, t)$ is trained to approximate $\nabla \log p_t(x)$. The reverse-time sampler substitutes this approximation into

$$\mathrm{d}X_t = \Big[f(X_t, t) - g(t)^2 s_\theta(X_t, t)\Big]\,\mathrm{d}t + g(t)\,\mathrm{d}\widetilde{W}_t, \qquad t : T \downarrow 0.$$

There are two distinct approximation errors: statistical error in the learned score and numerical error from discretizing the reverse dynamics.

The score can be learned without observing $\nabla \log p_t$ directly. When $X_0 \sim p_0$ and $X_t \sim q_t(\cdot \mid X_0)$, differentiation under the integral gives Fisher's identity

$$\nabla \log p_t(x) = \mathbb{E}[\nabla_x \log q_t(x \mid X_0) \mid X_t = x]\,.$$

Conditional expectation is an L^2 projection, so

$$\begin{aligned}&\mathbb{E}\,\|s_\theta(X_t, t) - \nabla_{x_t} \log q_t(X_t \mid X_0)\|_2^2 \\ &\qquad = \mathbb{E}\,\|s_\theta(X_t, t) - \nabla \log p_t(X_t)\|_2^2 + C_t,\end{aligned}$$

where C_t does not depend on θ. This denoising score-matching identity converts an inaccessible marginal-score regression problem into a supervised problem because the conditional score of the chosen corruption kernel is known analytically.

5.6 Variance Exploding (VE)

The VE process has zero drift and increasing diffusion, $\mathrm{d}X_t = g(t)\,\mathrm{d}W_t$. If $\sigma(t)^2 = \int_0^t g(s)^2\,\mathrm{d}s$, then its transition kernel is explicit:

$$X_t \mid X_0 = x_0 \sim \mathcal{N}(x_0, \sigma(t)^2 I), \qquad X_t = x_0 + \sigma(t)\varepsilon.$$

Consequently the conditional score is $\nabla_{x_t} \log p(x_t \mid x_0) = -(x_t - x_0)/\sigma(t)^2$, which provides the supervised target used in denoising score matching.

5.7 Variance Preserving (VP)

$$\mathrm{d}X_t = -\frac{1}{2}\beta(t) X_t\,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}W_t.$$

Writing $\overline{\alpha}(t) = \exp(-\int_0^t \beta(s)\,\mathrm{d}s)$ gives

$$X_t \mid X_0 = x_0 \sim \mathcal{N}\Big(\sqrt{\overline{\alpha}(t)}x_0, (1 - \overline{\alpha}(t))I\Big).$$

Unlike VE diffusion, the variance remains bounded and approaches one when the data have been standardized.

5.8 Discrete-Time Diffusion (DDPM)

Let $\beta_t \in (0,1)$, $\alpha_t = 1 - \beta_t$, and $\overline{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. The forward chain is

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} x_{t-1}, \beta_t I).$$

Gaussian closure yields the crucial one-step sampling identity

$$q(x_t \mid x_0) = \mathcal{N}(\sqrt{\overline{\alpha}_t} x_0, (1 - \overline{\alpha}_t) I), \quad x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t}\, \varepsilon.$$

Thus training need not simulate all preceding noising steps.

The exact posterior conditioned on x_0 is also Gaussian:

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\widetilde{\mu}_t(x_t, x_0), \widetilde{\beta}_t I), \qquad \widetilde{\beta}_t = \frac{1 - \overline{\alpha}_{t-1}}{1 - \overline{\alpha}_t} \beta_t.$$

The learned reverse kernel replaces the unavailable x_0 by a network prediction. In the common noise parameterization,

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \overline{\alpha}_t}} \varepsilon_\theta(x_t, t) \right).$$

5.9 Connection to VAE

Remark 5.9.1. Diffusion models can be interpreted as hierarchical VAEs: $p(x_0, \ldots, x_T) = p(x_T) \prod_{t=1}^{T} p(x_{t-1} | x_t)$.

Applying Jensen's inequality to the latent path $x_{1:T}$ gives a variational lower bound on $\log p_\theta(x_0)$. Its intermediate KL terms reduce to weighted denoising losses. Dropping known positive weights gives the widely used simplified objective

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, x_0, \varepsilon} \left\| \varepsilon - \varepsilon_\theta(x_t, t) \right\|_2^2.$$

This is not merely a heuristic reconstruction loss: it is a parameterization of multiscale denoising score matching.

5.10 Challenges with SDEs

Reverse-time sampling is a numerical SDE problem. Brownian paths are almost surely nowhere differentiable, so deterministic ODE error analysis cannot be applied unchanged. Euler–Maruyama has strong order 1/2 under standard global Lipschitz assumptions: over a fixed horizon, its pathwise root-mean- square error is typically $O(h^{1/2})$. Its weak error for sufficiently smooth test functions is often $O(h)$.

Dimension enters through both the score and the accumulated noise. A blanket claim such as $h = O(1/d)$ is not universal; the stable step size depends on the Lipschitz constant of

the reverse drift, the noise schedule, and the desired metric of accuracy. For a Gaussian target with precision matrix H, the Euler discretization of Langevin dynamics is stable only when $0 < h < 2/\lambda_{\max}(H)$. This simple example shows that stiffness, rather than dimension by itself, is the operative obstruction. Near low-noise times, the learned score may change rapidly and make the reverse equation stiff, which motivates nonuniform time grids and higher-order solvers.

5.11 Probability Flow ODE

For an SDE with state-independent scalar diffusion $g(t)$, define

$$\mathrm{d}X_t = \left[f(X_t, t) - \frac{1}{2} g(t)^2 \nabla \log p_t(X_t) \right] \mathrm{d}t.$$

Substitution into the continuity equation shows that this deterministic ODE and the forward SDE have the same one-time marginal densities p_t. The factor $1/2$ distinguishes the probability-flow ODE from the reverse SDE, whose score coefficient is $g(t)^2$.

Along an ODE trajectory, exact likelihoods satisfy

$$\log p_0(X_0) = \log p_T(X_T) + \int_0^T \nabla \cdot \left(f - \frac{1}{2} g^2 s_\theta \right)(X_t, t)\, \mathrm{d}t.$$

Remark 5.11.1. Sharing marginals does not mean sharing sample paths. The ODE gives a deterministic coupling between p_T and p_0, whereas the reverse SDE gives a stochastic coupling. The ODE permits adaptive deterministic solvers and likelihood evaluation, but may amplify score error differently from the SDE; neither route is uniformly faster or more stable for every model.

5.12 Fokker–Planck Equation

For a general Itô diffusion $\mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + G(X_t, t)\,\mathrm{d}W_t$ with $a = GG^\top$, the density evolves as

$$\partial_t p_t(x) = -\sum_i \partial_i \{ f_i(x, t) p_t(x) \} + \frac{1}{2} \sum_{i,j} \partial_{ij} \{ a_{ij}(x, t) p_t(x) \}.$$

This is the adjoint of the backward generator $\mathcal{L}_t \varphi = f \cdot \nabla \varphi + \frac{1}{2} \operatorname{tr}(a \nabla^2 \varphi)$. Indeed, Itô's formula gives $\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}[\varphi(X_t)] = \mathbb{E}[(\mathcal{L}_t \varphi)(X_t)]$; integration by parts moves the derivatives from φ onto p_t and produces the equation above. This weak derivation also clarifies the boundary conditions needed for probability mass to be conserved.

5.13 Summary of Diffusion Models

VE and VP processes are two choices of a forward corruption path, not two different principles of generation. Both expose conditional Gaussian scores, both support denoising

score matching, and both generate by solving reverse dynamics. DDPM is a discrete VP construction. Once a score has been learned, one may use a stochastic reverse SDE or the deterministic probability-flow ODE; the former and latter agree at the level of one-time marginals when the score is exact.

5.14 Score vs Likelihood-Based Learning

Maximum likelihood estimates parameters by matching normalized densities. In a regular correctly specified parametric model, its asymptotic covariance is the inverse Fisher information and is efficient among regular estimators. That conclusion does not make likelihood computationally available: an energy-based model may have an intractable partition function.

Score matching replaces density agreement by agreement of log-density gradients, so the partition function disappears after differentiation. The price is a different estimating equation whose asymptotic covariance need not attain the likelihood bound. Functional inequalities connect these two levels of analysis. Poincaré-type inequalities control fluctuations of estimating functions, while log-Sobolev inequalities control entropy decay of Langevin samplers. They therefore help quantify both statistical efficiency and computational mixing, although neither quantity alone determines the other.

5.15 Numerical Sampling Refinements

The reverse SDE and probability-flow ODE specify continuous dynamics, but an implementation sees only finitely many score-network evaluations. A sampler must allocate these evaluations across noise levels and choose how to correct local discretization error. The resulting algorithm can differ substantially from a naive Euler discretization even when both use the same trained score.

5.15.1 Predictor–Corrector Methods

A predictor advances the state from noise level t_k to t_{k-1} using one step of a reverse-SDE or ODE integrator. Holding t_{k-1} fixed, a corrector then applies one or more approximate MCMC transitions targeting $p_{t_{k-1}}$. A basic unadjusted Langevin corrector is

$$x^{(j+1)} = x^{(j)} + h, s_\theta(x^{(j)}, t_{k-1}) + \sqrt{2h}\,\xi_j, \qquad \xi_j \sim \mathcal{N}(0, I).$$

The predictor changes the target distribution along the diffusion path; the corrector reduces the lag relative to the current target. It does not literally "ensure equilibrium" after finitely many unadjusted steps, since score error and Langevin discretization both remain. This distinction matters when interpreting predictor–corrector sampling as approximate annealed MCMC.

5.16 Architecture and Implementation

5.16.1 Score Network

The network $s_\theta(x, t)$ must represent a vector field with the same shape as x and must respond to noise level t. A sinusoidal or Fourier embedding of t is passed through an MLP and injected into residual blocks. For images, a U-Net is natural: the contracting path builds low-resolution features with large receptive fields, the expanding path restores spatial resolution, and skip connections preserve fine detail that would otherwise be lost.

Self-attention blocks allow distant spatial locations to interact, while convolutional blocks encode local translation structure. The output is often parameterized as a score, noise ε, clean sample x_0, or velocity v. These parameterizations are algebraically convertible, but their target scales differ across time and therefore change optimization conditioning.

Example 5.16.1 (Converting noise prediction to a score)**.** For VP noising $x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\varepsilon$, the conditional score is

$$\nabla_{x_t} \log q(x_t \mid x_0) = -\frac{\varepsilon}{\sqrt{1-\overline{\alpha}_t}}.$$

Hence a noise predictor induces $s_\theta(x_t, t) = -\varepsilon_\theta(x_t, t)/\sqrt{1-\overline{\alpha}_t}$.

5.16.2 Practical Considerations

Discretization error is controlled jointly by the time grid and numerical solver. Euler–Maruyama is first order in the weak sense for SDEs, while Heun and modern ODE solvers can reduce error for probability-flow sampling. More function evaluations improve local accuracy but increase latency, so quality is usually plotted against the number of network evaluations rather than the nominal number of time steps.

The noise schedule determines signal-to-noise ratio $\mathrm{SNR}(t) = \overline{\alpha}_t/(1-\overline{\alpha}_t)$. Sampling t uniformly does not weight SNR levels uniformly, and an unweighted loss can be dominated by either nearly clean or nearly pure-noise examples. Time sampling and loss weights are therefore part of the statistical objective, not merely implementation details.

Conditional models provide side information c to the score network. In images, cross-attention lets spatial features query text embeddings. Latent diffusion first learns an autoencoder E, D and trains diffusion on $z = E(x)$; sampling returns $D(z_0)$. This lowers spatial dimension and cost, but reconstruction error and the geometry of the learned latent space become additional sources of bias.

5.17 Generalizing Diffusion Models

5.17.1 Observation Process View

Let $X \sim P$ be data. Define noisy observations: $Y_t = X + \text{noise}$

- As $t \to \infty$: Y_t becomes pure noise
- As $t \to 0$: recover original data

5.17.2 Stochastic Localization

$$p(X \mid Y_t)$$

defines evolving distributions that interpolate between prior and data.

5.18 Diffusion on Discrete Spaces

- Embed discrete data into continuous space
- Masking / token corruption (language models)
- Define discrete Markov processes directly

5.18.1 Continuous-Time Markov Chain

Generator matrix A: $A_{xy} \geq 0, \quad A_{xx} = -\sum_{y \neq x} A_{xy}$.

Generator: $\mathcal{L}f(x) = \sum_{y \neq x} A_{xy}(f(y) - f(x))$.

Remark 5.18.1. Defines jump process with exponential waiting times.

5.19 Reversing Markov Processes

Theorem 5.19.1. *Let (P_t) be a Markov process with generator A and marginals p_t. The reverse process has generator $B_t(x, y) = A_{yx}\frac{p_t(y)}{p_t(x)}, \quad x \neq y$.*

Remark 5.19.1. Discrete analogue of reverse SDE.

5.19.1 Example: Bit-Flip Process

On $\{-1, 1\}^n$:

- Forward: flip each coordinate randomly
- Reverse: depends on ratio $\frac{p_t(x^{(i)})}{p_t(x)}$

5.20 Stochastic Interpolants

Definition 5.20.1. A **stochastic interpolant** between distributions ρ_0, ρ_1 is: $x_t = I(t, x_0, x_1) + \gamma(t)z, \quad z \sim \mathcal{N}(0, I)$.

Example: $x_t = (1-t)x_0 + tx_1 + \sqrt{2t(1-t)}\, z$.

Remark 5.20.1. Generalizes diffusion models to arbitrary endpoints.

5.20.1 Dynamics

- ODE: $dx_t = b_t dt$
- SDE: $dx_t = (b_t + s_t)dt + \sqrt{2\epsilon} dW_t$

5.21 Schrödinger Bridge

5.21.1 Problem

Find a stochastic process matching: $X_0 \sim \rho_0, \quad X_T \sim \rho_1$ while minimizing KL divergence to a reference process.

Definition 5.21.1. $\pi^* = \arg\min_\pi \mathrm{KL}(\pi \| p_{\mathrm{ref}})$ subject to endpoint constraints.

For a Brownian reference with diffusivity ε, the dynamic form is

$$\inf_{(\rho_t, v_t)} \int_0^T \int \frac{1}{2} \|v_t(x)\|_2^2 \, \rho_t(x) \, \mathrm{d}x \, \mathrm{d}t$$

subject to

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = \frac{\varepsilon}{2} \Delta \rho_t, \qquad \rho_0, \rho_T \text{ fixed.}$$

The diffusion term distinguishes the Schrödinger problem from deterministic optimal transport. As $\varepsilon \downarrow 0$, the dynamic problem formally approaches the Benamou–Brenier formulation of quadratic optimal transport.

5.21.2 Connection to Optimal Transport

- Equivalent to entropy-regularized OT
- Balances transport cost and entropy

5.22 Iterative Proportional Fitting (IPF)

- Alternate between matching forward/backward marginals
- Converges to Schrödinger bridge solution

Remark 5.22.1. Key idea: alternating KL projections.

More precisely, the two endpoint constraints define convex sets of path measures. Each IPF step is an information projection onto one constraint set, implemented by rescaling with a function of the corresponding endpoint. The Pythagorean identity for KL projections makes the objective decrease monotonically; under positivity assumptions, the iterates converge to the unique Schrödinger bridge. In the static entropic optimal-transport problem, the same iteration is the Sinkhorn scaling algorithm.

5.23 Diffusion vs Schrödinger Bridge

Standard diffusion fixes a noising reference process and learns to reverse it. A Schrödinger bridge instead chooses, among all path measures connecting specified endpoint laws ρ_0 and ρ_1, the one closest in relative entropy to a reference diffusion. Diffusion generation is therefore a special endpoint-transport problem; the bridge formulation makes both endpoints and the path-space regularization explicit.

When the reference noise tends to zero, the bridge approaches deterministic optimal transport. At positive noise it spreads mass across paths, improves regularity, and permits stochastic control and likelihood-ratio tools, but it requires solving coupled forward and backward marginal constraints.

5.24 Flow Matching

Diffusion models learn a score and then insert it into a reverse SDE or probability-flow ODE. Flow matching learns the ODE velocity field directly. Let (X_0, X_1) be a coupling of a source distribution p_0 and target p_1, and define a differentiable interpolation

$$X_t = I_t(X_0, X_1), \qquad t \in [0, 1].$$

The conditional path velocity $\dot{X}_t = \partial_t I_t(X_0, X_1)$ depends on the unobserved endpoints. Its regression onto the current state defines the marginal velocity

$$v_t^*(x) = \mathbb{E}[\dot{X}_t \mid X_t = x].$$

Theorem 5.24.1 (Marginalization of conditional velocities)**.** *Assume the interpolation has smooth densities p_t and sufficient integrability. Then (p_t, v_t^*) satisfies the continuity equation*

$$\partial_t p_t + \nabla \cdot (p_t v_t^*) = 0.$$

Consequently, the ODE $\dot{Z}_t = v_t^*(Z_t)$ transports p_0 to p_t.

Proof. For a smooth compactly supported test function φ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\varphi(X_t) = \mathbb{E}\langle \nabla\varphi(X_t), \dot{X}_t\rangle = \mathbb{E}\langle \nabla\varphi(X_t), v_t^*(X_t)\rangle.$$

Writing both sides as integrals against p_t and integrating by parts gives the weak form of the continuity equation. □

The population flow-matching loss is

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t,X_0,X_1}\left\|v_\theta(X_t, t) - \dot{X}_t\right\|_2^2.$$

The L^2 projection property shows that its minimizer is v_t^*. Moreover, replacing the random conditional target by $v_t^*(X_t)$ changes the loss only by the irreducible conditional variance $\mathbb{E}\left\|\dot{X}_t - v_t^*(X_t)\right\|^2$, which is independent of θ.

Example 5.24.1 (Linear interpolation)**.** Let $X_0 \sim p_0$, $X_1 \sim p_1$ independently, and $X_t = (1-t)X_0 + tX_1$. Then $\dot{X}_t = X_1 - X_0$, so training requires only pairs of source and data samples. Although every conditional path is straight, the marginal velocity $\mathbb{E}[X_1 - X_0 \mid X_t = x]$ is generally nonlinear.

5.25 Stochastic Interpolants

A stochastic interpolant enriches the path with auxiliary noise:

$$X_t = I(t, X_0, X_1) + \gamma(t)Z, \qquad Z \sim \mathcal{N}(0, I),$$

where $\gamma(0) = \gamma(1) = 0$. Its conditional velocity target is

$$\dot{X}_t = \partial_t I(t, X_0, X_1) + \dot{\gamma}(t)Z.$$

Regressing this target on (X_t, t) again produces a velocity satisfying the continuity equation. By combining the velocity with the marginal score, one can construct an entire family of SDEs with the same marginals. This unifies deterministic flow matching, score-based diffusion, and Schrödinger-type bridges at the level of probability paths.

Example 5.25.1 (Gaussian endpoints)**.** If $X_0 \sim \mathcal{N}(m_0, C_0)$ and $X_1 \sim \mathcal{N}(m_1, C_1)$ are independent, then a linear stochastic interpolant remains Gaussian. Conditional Gaussian regression makes $v_t^*(x)$ affine in x, so the learned ODE reduces to a linear differential equation whose mean and covariance exactly match the interpolating Gaussian path.

5.26 Flow Maps and Few-Step Generation

For an ODE $\dot{X}_t = v_t(X_t)$, let $\Phi_{s,t}(x)$ denote the solution at time t initialized at x at time s. Flow maps satisfy the semigroup law

$$\Phi_{s,u} = \Phi_{t,u} \circ \Phi_{s,t}, \qquad s \le t \le u.$$

A numerical sampler approximates this map by many local solver steps. A few-step generator instead learns $F_\psi(x, s, t) \approx \Phi_{s,t}(x)$ directly, often by distilling a well-resolved teacher trajectory.

One consistency objective samples $s < t < u$ and penalizes

$$\left\| F_\psi(x, s, u) - F_{\bar{\psi}}(F_\psi(x, s, t), t, u) \right\|_2^2,$$

where $\bar{\psi}$ is a slowly updated target network. Boundary consistency requires $F_\psi(x, t, t) = x$. If these identities hold exactly on the relevant state distribution, recursive one-step and multistep evaluations agree.

Remark 5.26.1. Few-step generation trades discretization cost for approximation error in a learned global map. It can be much faster than integrating a score model, but errors are less locally correctable and may compound when a learned map is composed outside its training distribution.

5.27 Neural Samplers for Unnormalized Targets

Let $\pi(x) = e^{-U(x)}/Z$ be an unnormalized target and let $X = T_\theta(Z_0)$ with $Z_0 \sim q_0$. If the transport is invertible, its density q_θ is tractable and reverse-KL training gives

$$\mathrm{KL}(q_\theta \| \pi) = \mathbb{E}_{q_\theta}[\log q_\theta(X) + U(X)] + \log Z.$$

The unknown $\log Z$ is constant in θ, so samples from the target are not required. This is the basis of variational neural samplers and normalizing-flow proposals for energy-based models.

Reverse KL can ignore separated modes because it averages under q_θ. Forward KL is mass-covering but requires target samples or importance weights. Hybrid methods train a transport, correct it with MCMC, and recycle corrected samples. The learned map improves global movement while the Markov correction restores asymptotic exactness.

Example 5.27.1 (Transport-assisted independence sampler)**.** Use a flow density q_θ as an independence proposal in Metropolis–Hastings. From state x, propose $y \sim q_\theta$ and accept with probability

$$\alpha(x, y) = 1 \wedge \frac{e^{-U(y)} q_\theta(x)}{e^{-U(x)} q_\theta(y)}.$$

The normalizing constant cancels. A good flow yields high acceptance, while the MH step protects against approximation bias.

5.28 Schrödinger Bridges as Stochastic Optimal Control

Consider the controlled diffusion

$$\mathrm{d}X_t = (b(X_t, t) + u_t(X_t))\,\mathrm{d}t + \sqrt{2\varepsilon}\,\mathrm{d}W_t.$$

Girsanov's theorem identifies the path-space relative entropy between the controlled and reference processes:

$$\mathrm{KL}(P^u \| P^0) = \mathrm{KL}(\rho_0^u \| \rho_0^0) + \frac{1}{4\varepsilon} \mathbb{E}_{P^u} \int_0^T \|u_t(X_t)\|_2^2 \,\mathrm{d}t,$$

under the usual absolute-continuity and Novikov conditions. Thus a Schrödinger bridge is a minimum-energy stochastic-control problem with hard endpoint-distribution constraints.

Introducing endpoint potentials converts the constrained problem into a stochastic control problem. The optimal feedback is related to a value function V by $u_t^* = -2\varepsilon \nabla V_t$ under a common sign convention, and V satisfies a Hamilton–Jacobi–Bellman equation. The same potential appears in the Doob h-transform of the reference process.

Example 5.28.1 (Conditioning Brownian motion on an endpoint)**.** A Brownian motion conditioned on $X_T = y$ becomes a Brownian bridge with drift

$$u_t(x) = \frac{y - x}{T - t}.$$

The drift is weak early and diverges near T, forcing the endpoint. This is the point-mass analogue of steering an entire initial law to a terminal law.

5.29 Adjoint Methods and Adjoint Matching

Backpropagating through a finely discretized flow stores every intermediate state. Continuous adjoint sensitivity replaces this with a backward ODE. For

$$\dot{X}_t = v_\theta(X_t, t), \qquad J(\theta) = \Phi(X_T) + \int_0^T \ell(X_t, t)\,\mathrm{d}t,$$

define the adjoint $a_t = \nabla_{X_t} J$. It satisfies

$$-\dot{a}_t = (\nabla_x v_\theta(X_t, t))^\top a_t + \nabla_x \ell(X_t, t), \qquad a_T = \nabla\Phi(X_T),$$

and

$$\nabla_\theta J = \int_0^T (\nabla_\theta v_\theta(X_t, t))^\top a_t\,\mathrm{d}t.$$

This is obtained by differentiating the state equation, introducing a Lagrange multiplier, and integrating by parts so that state sensitivities cancel.

Adjoint matching methods learn a local model of the backward costate or value gradient from trajectory data, then use it as a regression target for the control or velocity field. The conceptual parallel to flow matching is important: flow matching regresses a forward conditional velocity, whereas adjoint matching regresses the backward signal specifying how future cost changes with the current state. In stochastic control the adjoint becomes a backward stochastic differential equation, and naive deterministic adjoints omit martingale terms.

Example 5.29.1 (Linear–quadratic control)**.** For $\dot{x} = Ax + Bu$ and cost $\frac{1}{2}x_T^\top Q_T x_T + \frac{1}{2}\int (x^\top Q x + u^\top R u)\,\mathrm{d}t$, the adjoint obeys $-\dot{a} = Qx + A^\top a$ and stationarity gives $u^* = -R^{-1}B^\top a$. With $a_t = P_t x_t$, these equations reduce to the Riccati differential equation.

5.30 Conditional Generation and Guidance

Let c denote a condition. Bayes' rule gives the exact score identity

$$\nabla_x \log p_t(x \mid c) = \nabla_x \log p_t(x) + \nabla_x \log p_t(c \mid x).$$

Classifier guidance estimates the first term with an unconditional diffusion model and the second with a noise-conditional classifier. Scaling the likelihood gradient by $w > 1$ sharpens adherence to the condition but samples from a tempered, altered distribution rather than the original conditional.

Classifier-free guidance trains conditional and null-conditioned predictions in one network. With score parameterization, a common convention is

$$s_{\mathrm{CFG}}(x, t, c) = s_\theta(x, t, \varnothing) + w\Big[s_\theta(x, t, c) - s_\theta(x, t, \varnothing)\Big].$$

At $w = 1$ this is the conditional score; larger w improves condition fidelity at a possible cost in diversity and calibration.

5.30.1 Inverse Problems and Inference-Time Steering

Suppose $Y = AX + \eta$ with $\eta \sim \mathcal{N}(0, \sigma_y^2 I)$. At the clean-data level,

$$\nabla_x \log p(y \mid x) = \frac{1}{\sigma_y^2} A^\top (y - Ax).$$

Posterior sampling requires the noisy-time likelihood score $\nabla_{x_t} \log p_t(y \mid x_t)$, which is generally intractable because it integrates over $X_0 \mid X_t$. Practical methods insert a denoised estimate $\widehat{x}_0(x_t)$ and differentiate $\log p(y \mid \widehat{x}_0(x_t))$, or alternate prior denoising with a data consistency step.

Example 5.30.1 (Inpainting)**.** If A selects observed pixels, then $A^\top(y - Ax)$ is zero on missing pixels and equals the residual on observed pixels. Guidance therefore enforces the measurement only where data exist, while the generative prior fills the unobserved region. Excessive guidance can create seams because the approximate likelihood score is evaluated away from the true posterior geometry.

Inference-time steering is broader than inverse problems: rewards, constraints, classifier logits, or differentiable simulators can supply a guidance gradient. Such steering is exact only when the added gradient is the score of a well-defined likelihood at the current noise level; otherwise it defines a useful controlled sampler but not necessarily the claimed Bayesian posterior.

5.31 Big Picture

The major constructions differ in what local object they learn. Score models learn $\nabla \log p_t$; flow matching learns a velocity v_t; flow-map distillation learns the finite-time transport $\Phi_{s,t}$; and stochastic control methods learn a feedback or value gradient. These objects are linked by the continuity and Fokker–Planck equations, but they induce different training targets, numerical errors, and opportunities for conditioning.

The research frontier combines them: a neural transport proposes global movement, an SDE or MCMC correction controls distributional error, an adjoint propagates downstream objectives, and guidance modifies the dynamics for a specific observation or reward.

5.32 Modern Deep Generative Modeling

5.32.1 Unifying View

We consider the problem: Learn $p_{\text{data}}(x)$ from samples.

The distinction between "likelihood-based" and "likelihood-free" methods is useful but incomplete. Flows evaluate an exact normalized likelihood; VAEs optimize a lower bound; energy models and NCE manipulate unnormalized density ratios; GANs compare samples through a discriminator; and diffusion models learn derivatives of log densities along a noising path. Each replaces a difficult global density-learning problem by a different tractable local object.

5.33 Generative Adversarial Networks (GANs)

5.33.1 Formulation

$$\min_{g\in\mathcal{G}} \max_{f\in\mathcal{F}} \left[\mathbb{E}_{x\sim p_{\text{data}}} \log f(x) + \mathbb{E}_{x\sim p_g} \log(1 - f(x))\right]$$

- g: generator, pushes $z \sim p_z$ to $x = g(z)$
- f: discriminator, estimates $P(y = 1|x)$

5.33.2 Optimal Discriminator

$$f^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}.$$

Proof. For a fixed generator, the objective is the integral of $p_{\text{data}}(x) \log f(x) + p_g(x) \log(1 - f(x))$. It can be maximized pointwise. For nonnegative a, b, strict concavity of $a \log u + b \log(1 - u)$ gives the maximizer $u = a/(a + b)$. □

5.33.3 Divergence Interpretation

Let $m = (p_{\text{data}} + p_g)/2$. Substitution of f^* gives

$$\begin{aligned} V(g, f^*) &= -\log 4 + \mathrm{KL}(p_{\text{data}}\|m) + \mathrm{KL}(p_g\|m) \\ &= -\log 4 + 2\,\mathrm{JS}(p_{\text{data}}, p_g). \end{aligned}$$

Hence the population optimum is attained exactly when $p_g = p_{\text{data}}$.

Remark 5.33.1. If the two distributions lie on disjoint low-dimensional supports, the Jensen–Shannon divergence is locally constant and the ideal discriminator saturates. This explains the weak generator gradients that motivate alternative losses and Wasserstein GANs.

5.33.4 Wasserstein GAN

$$\min_{g} \max_{f\in\mathrm{Lip}(1)} \left[\mathbb{E}_{p_{\text{data}}} f(x) - \mathbb{E}_{p_g} f(x)\right]$$

- Approximates $W_1(p_{\text{data}}, p_g)$
- Requires Lipschitz constraint on f

Remark 5.33.2. More stable than standard GANs when supports are disjoint.

Theorem 5.33.1 (Kantorovich–Rubinstein duality)**.** *For probability measures with finite first moment,*

$$W_1(P, Q) = \sup_{\|f\|_{\mathrm{Lip}}\le 1} \{\mathbb{E}_P f(X) - \mathbb{E}_Q f(X)\}.$$

Thus the WGAN critic estimates a meaningful transport distance even when the supports are disjoint.

5.33.5 Practical Issues

Mode collapse occurs when distinct latent regions map to a small collection of outputs that fool the current discriminator. The generator and discriminator also solve a nonconvex–nonconcave game, so simultaneous gradient descent need not approach equilibrium and may rotate around it. If the discriminator is too weak, it supplies an uninformative discrepancy; if it is optimized too aggressively, it can saturate and leave the generator with vanishing or highly variable gradients. Regularization, alternative generator losses, gradient penalties, and multiple discriminator updates address different parts of this problem but do not provide a general convergence guarantee.

5.34 Noise Contrastive Estimation (NCE)

5.34.1 Idea

Reduce density estimation to classification:

$$\ell(f) = \sum_{i=1}^{N} \log f(x_i^{\text{data}}) + \sum_{j=1}^{M} \log(1 - f(x_j^{\text{noise}})).$$

- Data: $x \sim p_{\text{data}}$
- Noise: $x \sim q$

5.34.2 Optimal Classifier

Suppose there are k noise observations for each data observation. Bayes' rule gives

$$f^*(x) = \mathbb{P}(Y = 1 \mid X = x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + kq(x)}.$$

Taking odds recovers the density ratio:

$$\frac{p_{\text{data}}(x)}{q(x)} = k\frac{f^*(x)}{1 - f^*(x)}.$$

For an unnormalized model $p_\theta(x) = \exp(s_\theta(x) - c)$, NCE treats the log-normalizer c as an ordinary parameter and fits the logistic logit $s_\theta(x) - c - \log(kq(x))$. Under identifiability and standard uniform-law conditions, the population classifier is unique, so the NCE estimator is consistent.

Remark 5.34.1. Recovers density ratio $\frac{p_{\text{data}}}{q}$.

5.34.3 Key Insight

NCE avoids explicit partition-function evaluation because classification identifies a density ratio. Its success nevertheless depends on overlap: when the noise distribution q is easily separable from the data, the classifier saturates and the ratio is poorly estimated in the regions that matter. This is another manifestation of the curse of dimensionality.

5.34.4 Telescoping Density Ratio Estimation

Construct overlapping intermediate distributions $p_0, \dots, p_m$ and use

$$\log \frac{p_0(x)}{p_m(x)} = \sum_{j=0}^{m-1} \log \frac{p_j(x)}{p_{j+1}(x)}.$$

Each adjacent ratio is learned by a separate classification problem. The telescoping construction trades one difficult high-dimensional discrimination problem for several easier local ones, at the cost of accumulated estimation error.

5.35 Evaluation Metrics

5.35.1 Inception Score (IS)

$$\mathbb{E}_{x\sim p_g}\mathrm{KL}(p(y|x)\|p(y))$$

- High if:
  - confident predictions (low entropy $p(y|x)$)
  - diverse outputs (high entropy $p(y)$)

Remark 5.35.1. Can be gamed (memorization).

5.35.2 Fréchet Inception Distance (FID)

For Gaussian feature approximations $\mathcal{N}(\mu, \Sigma)$ and $\mathcal{N}(\mu', \Sigma')$, the squared Wasserstein distance is

$$\mathrm{FID} = \|\mu - \mu'\|_2^2 + \mathrm{tr}\left(\Sigma + \Sigma' - 2(\Sigma^{1/2}\Sigma'\Sigma^{1/2})^{1/2}\right).$$

FID compares Gaussian approximations to two feature distributions. The mean term detects systematic feature displacement, while the covariance term responds to spread and correlations; neither term sees non-Gaussian structure beyond the selected feature representation.

Remark 5.35.2. Empirical FID is biased at finite sample sizes and depends on the feature extractor. It is not a likelihood and can fail to detect memorization or within-class mode dropping.

5.36 Diffusion Models (Modern View)

5.36.1 Score-Based Perspective

The score $s_t(x) = \nabla \log p_t(x)$ is learned by denoising score matching and inserted into Langevin dynamics, a reverse SDE, or the probability-flow ODE. Unlike a normalized density, the score is local and invariant to an unknown multiplicative normalizer.

5.36.2 Forward / Reverse Processes

The forward process destroys information until X_T is approximately Gaussian. The reverse process reconstructs the sequence of intermediate marginals using the learned score. Errors at high noise affect global structure; errors near the data endpoint affect fine detail.

5.36.3 Two-Step View

Training estimates $s_\theta(x, t)$ over the full noise range. Generation then solves dynamics using that fixed network. This separation explains why a new numerical solver or guidance rule can be applied at inference time without retraining the base score model.

5.37 Probability Flow and Sampling Refinements

The reverse SDE, deterministic probability-flow ODE, and predictor–corrector samplers share the same learned score but have different path laws and discretization behavior. The ODE supports deterministic likelihood evaluation; the SDE preserves stochasticity; a corrector adds local MCMC steps to reduce off-manifold numerical error.

5.38 Score Matching on Discrete Spaces

Define discrete gradient: $\nabla f(x) = (f(x^{(i)}) - f(x))_{i=1}^d$.

Alternative objective: $L_p(q) = \sum_{i=1}^d \mathbb{E}_{x\sim p} \log q(x_i|x_{-i})$.

Remark 5.38.1. Pseudo-likelihood is discrete analogue of score matching.

5.39 Statistical Efficiency

5.39.1 MLE vs Score Matching

Under correct specification and regularity, maximum likelihood attains the inverse Fisher-information covariance and is asymptotically efficient. Score matching replaces the likelihood score with spatial derivatives, avoiding the normalizer but generally producing a different estimating equation and a larger sandwich covariance. Computational tractability is purchased with a possible statistical-efficiency loss.

5.39.2 Functional Inequalities

Poincaré inequalities control variance by Dirichlet energy and yield L^2 mixing. Log-Sobolev inequalities control relative entropy by Fisher information and yield exponential KL convergence. In score estimation, the same inequalities compare Fisher divergence to stronger distributional metrics, linking estimation error to sampling error.

5.40 Final Big Picture

Generative modeling is distribution learning under operational constraints. Optimization-based methods, Markov samplers, transports, and learned dynamics are not isolated paradigms: modern systems routinely combine them. The central tradeoffs are statistical efficiency, computational tractability, exactness, coverage, and latency. Which approximation is appropriate depends on the downstream task, not only on visual sample quality.

Chapter 6

Calibration and Distribution-Free Uncertainty

6.1 Calibration

A probabilistic predictor $f : \mathcal{X} \to [0, 1]$ is calibrated when its reported probabilities agree with empirical frequencies at the population level:

$$\mathbb{E}[Y \mid f(X)] = f(X) \qquad \text{almost surely.}$$

The frequently written condition $\mathbb{P}(Y = 1 \mid f(X) = p) = p$ must be interpreted through a regular conditional probability because the event $\{f(X) = p\}$ usually has probability zero. Calibration concerns the reliability of uncertainty reports, not their resolution or predictive accuracy.

Example 6.1.1 (Calibration does not imply usefulness)**.** If $\mathbb{P}(Y = 1) = 1/2$, the constant forecast $f(x) = 1/2$ is calibrated but does not distinguish easy observations from difficult ones. Conversely, a highly accurate classifier can be miscalibrated if it reports probability 0.99 on predictions that are correct only 0.8 of the time.

6.1.1 Calibration in Deep Learning

Cross-entropy is a proper population loss, but finite-sample optimization, model misspecification, distribution shift, early stopping, and selection on the same validation data can all produce miscalibration. In separable classification, unregularized logistic loss can drive logit norms toward infinity even after the decision boundary has stabilized, creating extreme probabilities. Post-training objectives may further change logit scale without preserving probabilistic meaning.

Calibration must therefore be evaluated on held-out data drawn from the deployment distribution. Under covariate or label shift, a calibrator learned on the training distribution need not remain calibrated.

6.2 Proper Scoring Rules

A binary scoring rule $S(q, y)$ is *proper* if, whenever $Y \sim \text{Bernoulli}(p)$,

$$p \in \arg\max_{q \in [0,1]} \mathbb{E}_{Y \sim \text{Bernoulli}(p)} S(q, Y).$$

It is *strictly proper* when $q = p$ is the unique maximizer. Properness is an incentive-compatibility property: a forecaster maximizes expected score by reporting its true belief.

Examples:

- Log score: $S(q, y) = y \log q + (1 - y) \log(1 - q)$.
- Brier score: $S(q, y) = -(q - y)^2$.

Example 6.2.1 (Strict propriety of the Brier score)**.** For $Y \sim \text{Bernoulli}(p)$,

$$\mathbb{E}(q - Y)^2 = (q - p)^2 + p(1 - p).$$

The second term is independent of q, and the first is uniquely minimized at $q = p$.

Remark 6.2.1. Proper scoring rules incentivize truthful probabilities.

6.3 Measuring Calibration

6.3.1 Expected Calibration Error (ECE)

$$\text{ECE}(f) = \mathbb{E}\Big[\, |\mathbb{E}[Y \mid f(X)] - f(X)| \,\Big].$$

$$\text{ECE}_2(f) = \mathbb{E}\Big[\, (\mathbb{E}[Y \mid f(X)] - f(X))^2 \,\Big].$$

Remark 6.3.1. In practice, estimated via binning (reliability diagrams).

Given bins $B_1, \ldots, B_K$ partitioning $[0, 1]$, the common plug-in estimator is

$$\widehat{\text{ECE}} = \sum_{k=1}^{K} \frac{n_k}{n} \left| \frac{1}{n_k} \sum_{i : f_i \in B_k} Y_i - \frac{1}{n_k} \sum_{i : f_i \in B_k} f_i \right|.$$

This statistic estimates a *binned* calibration error, not the population ECE itself. Coarse bins hide local deviations; fine bins have high variance; the absolute value creates upward finite-sample bias. Comparing models with different binning schemes can therefore be misleading.

Example 6.3.1 (Cancellation within a bin)**.** Suppose half the observations in one bin receive forecast 0.2 with true frequency 0, and half receive forecast 0.8 with true frequency 1. The bin-level average forecast and outcome frequency are both 0.5, so binned ECE reports zero even though both subgroups are miscalibrated by 0.2.

6.3.2 Decomposition of Error

Let $C = \mathbb{E}[Y \mid f(X)]$. Conditional expectation is the L^2 projection onto the sigma-field generated by $f(X)$. Hence

$$\mathbb{E}[(f(X) - Y)^2] = \underbrace{\mathbb{E}[(f(X) - \mathbb{E}[Y|f(X)])^2]}_{\text{calibration error}} + \underbrace{\mathbb{E}[(\mathbb{E}[Y|f(X)] - Y)^2]}_{\text{intrinsic noise}}.$$

The cross term vanishes because $\mathbb{E}[Y - C \mid f(X)] = 0$. This is a Pythagorean identity, not an approximation.

6.4 Calibration Algorithms

Define: $\text{calib}(p) = \mathbb{P}(Y = 1 \mid f(X) = p)$.

Construct calibrated predictor: $\hat{f}(x) = \text{calib}(f(x))$.

Theorem 6.4.1. *$\hat{f}$ is calibrated and improves squared loss: $L(\hat{f}) = L(f) - \text{ECE}_2(f)$.*

Proof. Since $\hat{f}(X) = \mathbb{E}[Y \mid f(X)]$ is measurable with respect to $f(X)$,

$$\mathbb{E}[Y \mid \hat{f}(X)] = \mathbb{E}[\mathbb{E}[Y \mid f(X)] \mid \hat{f}(X)] = \hat{f}(X),$$

so $\hat{f}$ is calibrated. The preceding orthogonal decomposition gives $L(f) = L(\hat{f}) + \mathbb{E}[(f(X) - \hat{f}(X))^2]$; the last term is $\text{ECE}_2(f)$. □

6.4.1 Practical Methods

Histogram calibration replaces scores in a bin by the empirical outcome frequency. It is simple but discontinuous and sensitive to bin boundaries. Isotonic regression fits a nondecreasing calibration map by minimizing squared error; the pool-adjacent-violators algorithm computes the exact empirical solution. Monotonicity preserves ranking but may overfit when the calibration sample is small.

Temperature scaling assumes a trained multiclass logit vector $z(x)$ and fits one positive scalar T on held-out data:

$$\widehat{p}_k(x; T) = \frac{\exp(z_k(x)/T)}{\sum_j \exp(z_j(x)/T)}.$$

It changes confidence without changing the predicted class. Vector and matrix scaling are more expressive but require more calibration data. Platt scaling fits an affine transformation of a binary score before applying the logistic link.

Example 6.4.1 (Effect of temperature)**.** For binary logit $z = 4$, the unscaled probability is approximately 0.982. With $T = 2$, it becomes $\text{logistic}(2) \approx 0.881$. Thus $T > 1$ softens an overconfident model, whereas $T < 1$ sharpens it.

6.5 Issues in Continuous Case

When $f(X)$ has a continuous distribution, exact score levels are not observed repeatedly. Estimating $p \mapsto \mathbb{E}[Y \mid f(X) = p]$ is a one-dimensional nonparametric regression problem. Kernel or local-polynomial smoothing avoids hard bin boundaries but introduces bandwidth bias and boundary effects.

Calibration functionals can also be statistically irregular. An arbitrarily small perturbation of scores can split or merge level sets of $f(X)$, changing the conditional expectation that defines ECE. One alternative measures the distance to the set $\mathcal{C}$ of calibrated predictors:

$$d_{\mathrm{CE}}(f) = \inf_{g \in \mathcal{C}} \mathbb{E}|f(X) - g(X)|.$$

Another uses smooth test functions h and studies calibration moments

$$\mathbb{E}[(Y - f(X))h(f(X))].$$

Exact calibration makes every such moment zero. Restricting h to a bounded function class yields estimable kernel or adversarial calibration errors and connects calibration testing to integral probability metrics.

Example 6.5.1 (Kernel calibration moment)**.** With a positive-definite kernel k, consider

$$\mathbb{E}[(Y - f(X))(Y' - f(X'))k(f(X), f(X'))],$$

where the primed pair is independent. Under suitable characteristic kernels, this quantity vanishes only when the conditional residual mean is zero. Its empirical version is a U-statistic and avoids choosing histogram bins.

6.6 Extensions

6.6.1 Multiclass Calibration

For a probability vector $f(X) \in \Delta^{K-1}$, strong calibration requires

$$\mathbb{P}(Y = k \mid f(X)) = f_k(X), \qquad k = 1, \ldots, K.$$

Classwise calibration conditions only on $f_k(X)$ and is weaker; top-label calibration considers only the predicted class and its reported confidence. These notions answer different operational questions and should not be used interchangeably.

6.6.2 Multicalibration

Let $\mathcal{G}$ be a collection of potentially overlapping subpopulations. A predictor is approximately multicalibrated if its residual mean is small within every sufficiently populated group and score range. A representative condition is

$$|\mathbb{E}[Y - f(X) \mid X \in G,\ f(X) \in B]| \le \alpha$$

for all $G \in \mathcal{G}$ and relevant bins B. The guarantee is stronger than aggregate calibration and can expose systematic subgroup errors hidden by population averages. Statistical complexity depends on the richness of $\mathcal{G}$; uniform guarantees require capacity control and enough samples in each audited slice.

6.6.3 Decision Calibration

Calibration is ultimately useful because probabilities support decisions. For action set $\mathcal{A}$ and utility $u(a, y)$, a forecast q induces the Bayes action $a(q) \in \arg\max_a \mathbb{E}_{Y\sim q} u(a, Y)$. Decision calibration asks whether the predicted and realized utilities agree for the actions the model causes the user to take. It is weaker than full distribution calibration but can be better aligned with a specified downstream task.

Example 6.6.1 (Treatment threshold)**.** Treat when the predicted event probability exceeds a cost-dependent threshold c. Accurate calibration far away from c may have little decision value, whereas a small local bias near c can reverse many treatment decisions. Task-aware calibration emphasizes this relevant region.

6.7 Conformal Prediction

6.7.1 Goal

Point predictions conceal ambiguity. Conformal prediction wraps a fitted model in a set-valued predictor $T : \mathcal{X} \to 2^{\mathcal{Y}}$ with finite-sample marginal coverage

$$\mathbb{P}\{Y_{n+1} \in T(X_{n+1})\} \geq 1 - \delta.$$

The basic guarantee assumes exchangeability, not a correct regression model, homoskedastic noise, or asymptotics. Marginal coverage averages over test features and should not be read as coverage conditional on every $X = x$.

6.7.2 Nonconformity Score

A nonconformity score $s(x, y)$ measures how incompatible a candidate label is with the fitted model. In regression, the absolute residual $s(x, y) = |y - \widehat{m}(x)|$ produces constant-width intervals. If a scale model $\widehat{\sigma}(x)$ is available, the normalized residual

$$s(x, y) = \frac{|y - \widehat{m}(x)|}{\widehat{\sigma}(x)}$$

adapts interval width to heteroskedasticity.

For classification with estimated probabilities $\widehat{p}_k(x)$, the simple score $1 - \widehat{p}_y(x)$ includes labels whose probability exceeds a calibrated threshold. Adaptive prediction sets instead order classes by probability and score a label by the cumulative mass of classes ranked ahead of it. This tends to produce small sets on easy inputs and larger sets on ambiguous inputs.

6.7.3 Algorithm

In split conformal prediction, first fit the underlying predictor on a training sample independent of the n calibration observations. Compute $S_i = s(X_i, Y_i)$ on the calibration sample and let

$$k = \lceil (n+1)(1-\delta) \rceil, \qquad \tau = S_{(k)},$$

where $S_{(k)}$ is the kth order statistic, with $S_{(n+1)} = +\infty$ when $k = n+1$. Output $T(x) = \{y : s(x,y) \le \tau\}$.

For absolute residual scores in regression this becomes the interval $[\widehat{m}(x) - \tau, \widehat{m}(x) + \tau]$.

Example 6.7.1 (Numerical split-conformal interval)**.** Suppose $n = 99$ calibration residuals are available and the target miscoverage is $\delta = 0.1$. Then $k = \lceil 100(0.9) \rceil = 90$. If the 90th ordered residual is 2.3, the reported interval at a new feature vector is $[\widehat{m}(x) - 2.3, \widehat{m}(x) + 2.3]$. The use of $n+1$, rather than n, is the finite-sample correction that guarantees coverage.

6.7.4 Guarantee

Theorem 6.7.1. *If the calibration observations and the test observation are exchangeable, then*

$$\mathbb{P}\{Y_{n+1} \in T(X_{n+1})\} \ge 1 - \delta.$$

If the scores have no ties, the coverage is at most $1 - \delta + 1/(n+1)$.

Proof. Conditional on the unordered collection of $n+1$ scores, exchangeability makes the rank of the test score uniform on $\{1, \dots, n+1\}$. The test point is covered whenever its rank is at most k. Therefore the coverage is $k/(n+1) \ge 1-\delta$. Without ties, $k/(n+1) < 1-\delta+1/(n+1)$. □

6.7.5 High-Probability Version

The marginal theorem averages over the random calibration sample; it does not assert conditional coverage for every realized sample or every value of X. A distribution-free high-probability statement about the population score CDF follows from the Dvoretzky–Kiefer–Wolfowitz inequality:

$$\mathbb{P}\left(\sup_t |\widehat{F}_n(t) - F(t)| > \epsilon \right) \le 2e^{-2n\epsilon^2}.$$

Thus, with probability at least $1 - \gamma$, empirical and population score quantiles differ in probability level by at most $\sqrt{\log(2/\gamma)/(2n)}$. This statement should not be confused with distribution-free conditional coverage, which is impossible without further assumptions or weakened targets.

6.7.6 Full Conformal and Cross-Conformal Variants

Full conformal prediction refits the model after augmenting the training data with each candidate test label and computes a rank-based conformal p-value. It uses data efficiently and

retains exact exchangeable coverage, but it can be computationally prohibitive. Split conformal fits once and calibrates once, trading some statistical efficiency for speed. Jackknife+, CV+, and related cross-fitting methods reuse data across folds while controlling the dependence created by repeated fitting.

6.7.7 Distribution Shift

Ordinary exchangeability fails under covariate shift. If the conditional law $P(Y \mid X)$ is stable and the test-to-training density ratio $w(x) = \mathrm{d}P^X_{\text{test}} / \mathrm{d}P^X_{\text{train}}$ is known, weighted conformal ranks can restore test-distribution coverage. Estimating w adds error and can be unstable when the two feature distributions have poor overlap. Arbitrary concept shift cannot be repaired without new assumptions or labels from the deployment environment.

6.8 Key Insight

Calibration and conformal prediction solve different problems. Calibration asks whether reported probabilities have the right long-run meaning. Conformal prediction asks whether a data-dependent set covers the next response at a specified marginal rate. A calibrated classifier need not produce efficient conformal sets, and a conformal wrapper does not make the underlying class probabilities calibrated.

The phrase "distribution-free" also has a precise scope: the conformal rank argument is valid for every exchangeable joint distribution, but the resulting guarantee is marginal. Strong conditional, subgroup, or shifted-distribution guarantees require additional structure.

6.9 Big Picture

Probabilistic machine learning has three distinct layers. Inference methods such as MCMC and variational inference approximate a model distribution. Calibration checks whether predictive probabilities agree with observed frequencies. Conformal prediction adds a finite-sample coverage guarantee to a chosen score. None of these layers automatically validates the others: an exact Bayesian computation can be miscalibrated under model misspecification, and a conformal set can cover correctly while being too wide to be useful.